

King Saud University
College of Computer and Information Science
Information Technology Department

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Phase 4: Prototype

Project Name: راع - Ra'i



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Introduction

With the rapid advancement of technology and the widespread use of social media and online games, the time children spend on electronic devices has become a significant challenge for families. This phenomenon has led to the rise of digital addiction, making it increasingly difficult to control the amount of time children spend in front of screens, which has negatively impacted family bonds. Family moments have diminished significantly, and each member has become immersed in their own world, leading to a lack of deep family interaction.

Moreover, parents face difficulty in monitoring the content their children consume, raising concerns about the effects on their mental and emotional development. The busyness of parents adds complexity to this issue, especially considering how attached children are to their devices.

1. Problem Statement

The increased use of digital devices by children and their exposure to excessive or inappropriate content online poses a threat to their mental and emotional well-being, as well as reducing family interaction. Although parents wish to monitor and protect their children, limited time and daily pressures make it difficult to control their screen time.

This study aims to address this problem by developing an app that helps families organize children's screen time and provides flexible and effective solutions to reduce the impact of digital addiction.

- **Objectives:**
 - Provide an effective tool to organize children's screen time by setting specific usage times.
 - Immediate device lockdown during specific times, such as sleep time, prayer time, and family moments.
 - Track content and applications used by children, offering precise monitoring for parents.
 - Enhance family moments by reducing screen time and promoting interaction among family members.
- **User Needs:**
 - Parents need an **effective tool** to reduce the stress of monitoring children and ensuring their digital safety and the time they are spending on the screens.
 - Children need a **safe digital environment** that allows them to enjoy games and apps without exposure to inappropriate content.
 - Both parties need a **flexible tool** that allows them to engage with technology in a healthy way.

2. Methodology

"Ra'i" focuses on creating a user-friendly app to help parents manage their children's screen time effectively. It begins with defining clear goals and conducting target audience research to understand user needs, since our target audience are parents, we conducted the questions based on their view and concerns. We employed a combination of qualitative and quantitative methods through interviews and questionnaires.

To gain a deeper understanding of parents' perspectives and behaviours, we chose a sample of struggling parents whom we've noticed will enrich us with their point of view. We conducted structured interviews with open-ended questions. The interview questions were divided into three categories Personal Information, General Questions and Focused Questions. These interviews provided qualitative data that highlighted recurring pain points, such as difficulty enforcing time limits and the need for a motivational system, which directly informed the development of "Ra'i".

To complement the interviews, we designed a questionnaire to gather quantitative data from parents about their habits and challenges in managing their children's screen time. The questionnaire was designed for parents using Google Forms, so we distributed it to a group of concerned families via WhatsApp to complete. The questionnaire consisted of eight closed-ended questions that their responses helped identify common patterns and priorities, ensuring the app's features were tailored to address the most critical needs.

3. Result

a) Interview Results:

The interviews revealed that most parents are concerned about the negative effects of excessive screen time on their children's behaviour and health. They agreed that children who spend long hours on devices tend to have reduced focus, disrupted sleep patterns.

The data show that most respondents feel frustrated when trying to balance their children's screen time with family activities. Also, screen time can cause contentious issues between parents and kids. Many of the parents we spoke to mentioned that existing solutions for managing screen time are either too complex to set up and use, or do not include the in-depth reporting needed to be truly helpful. Parents said the need for customizable restrictions was important to them, as the ability to set specific hours for device-use and block certain applications would be helpful in finding a healthier balance for their children.

b) Questionnaire Results:

With a total of 77 responses, the survey results highlight significant concerns among parents regarding their children's screen time. 48% of respondents admitted they face difficulties in managing their children's device usage, while 55.8% expressed concern about not knowing which apps their children spend the most time on. Additionally, 41.6% of parents reported that their children spend 4-6 hours daily on screens, primarily gaming, watching videos, and browsing the internet.

When asked about the most critical times to limit screen usage, the majority pointed to bedtime, family time, study time, and meal times as essential moments for reducing distractions. However, the biggest challenges parents face are children resisting restrictions (32.5%) and lack of time to monitor usage (33.8%).

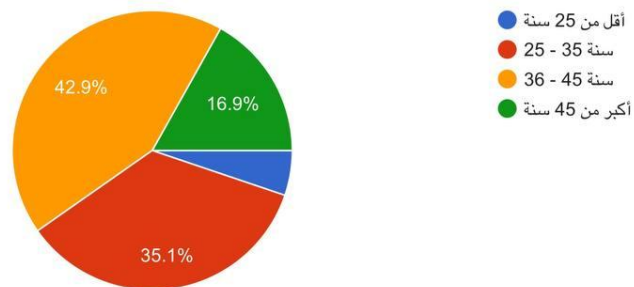
Regarding features they seek in a screen time management app, 64.9% emphasized the need for scheduled device locking, while 14.3% preferred detailed app usage reports. Moreover, 68.8% showed strong interest in a task-based reward system, where children earn screen time by completing homework or chores.

These insights reinforce the demand for an automated, flexible, and conflict-free solution like Ra'i, helping families create a balanced and healthier digital routine.

عمر ولي الأمر؟

77 responses

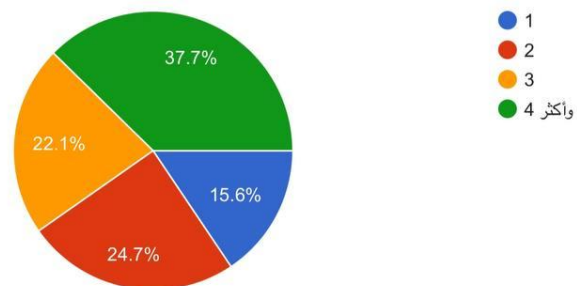
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كم عدد أطفالك؟

77 responses

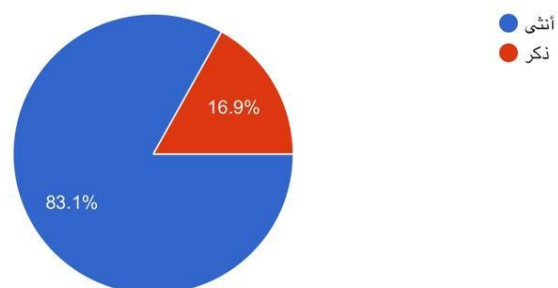
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جنس ولي الأمر؟

77 responses

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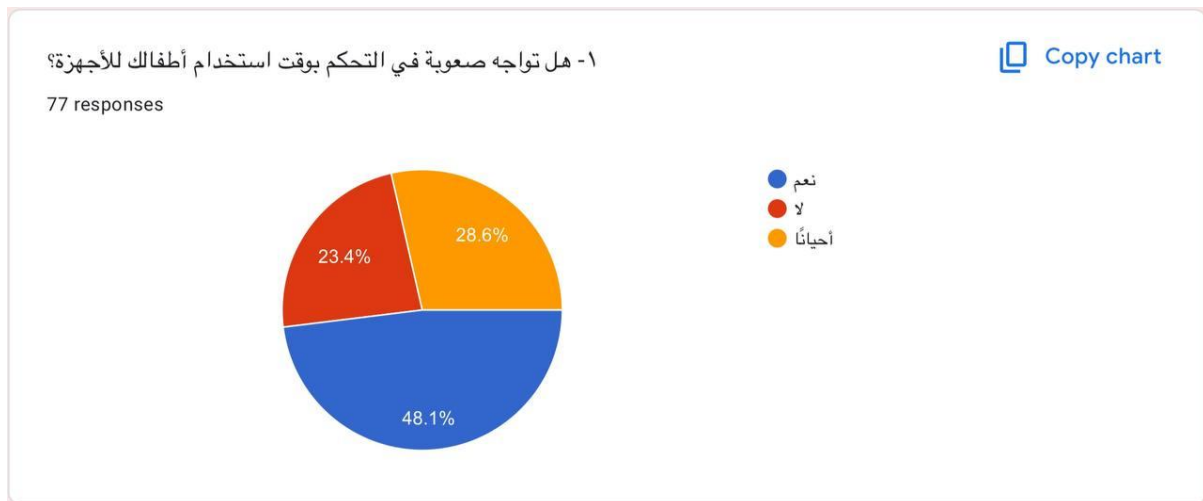
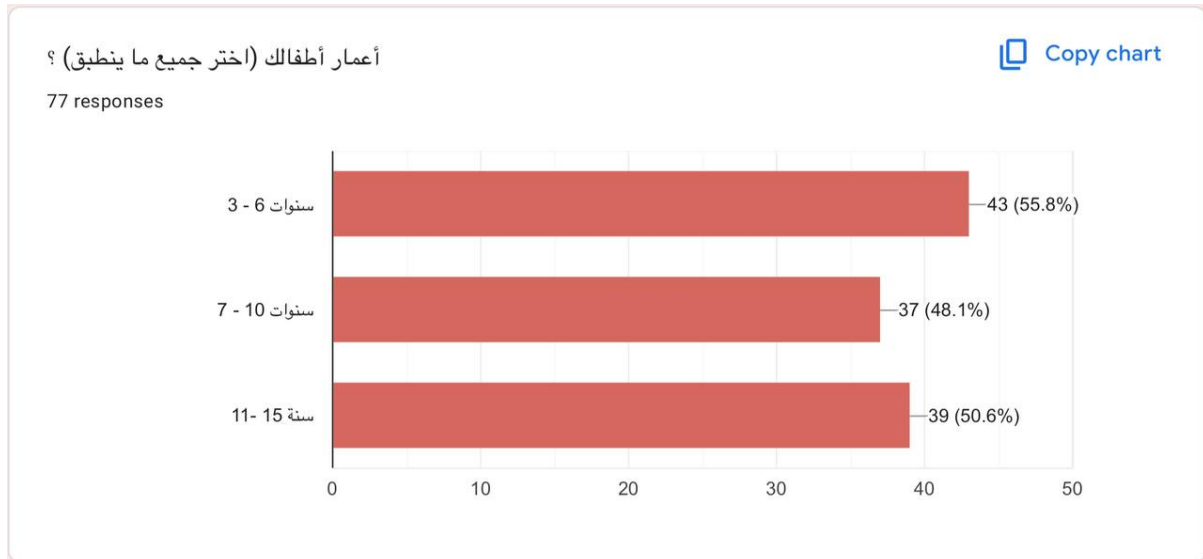


Figure 1 : Results of question 1

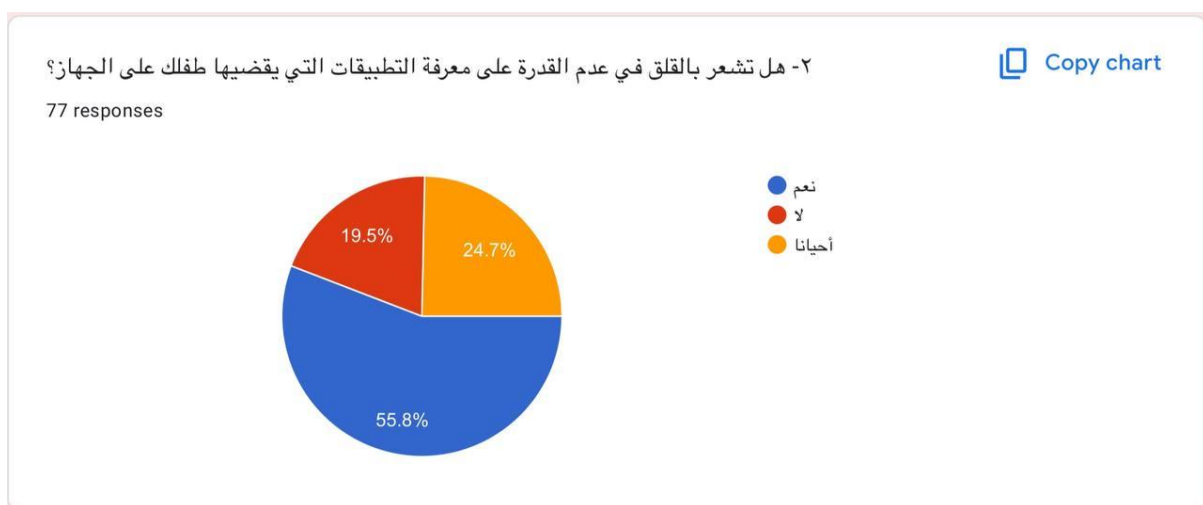


Figure 2 : Results of question 2

٣- كم عدد الساعات التي يقضيها أطفالك على الأجهزة يوميًا (تقريبًا)؟

77 responses

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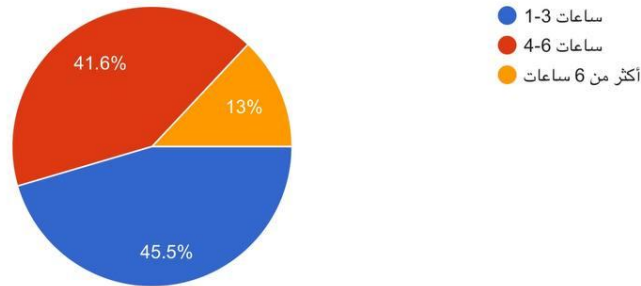


Figure 3 : Results of question 3

٤- ما أكثر الأنشطة التي يقوم بها أطفالك على الأجهزة؟

77 responses

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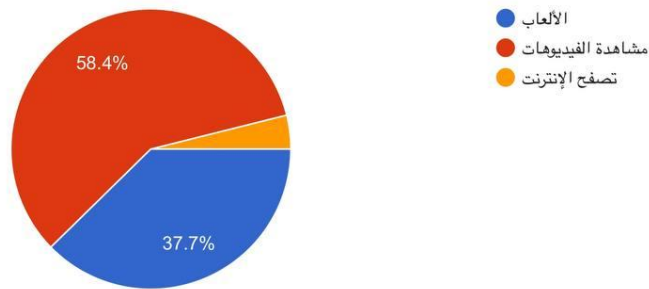


Figure 4 : Results of question 4

٥- ما هي أكثر الأوقات التي ترغب في تقليل استخدام أطفالك للأجهزة فيها (اختر جميع ما ينطبق)؟

77 responses

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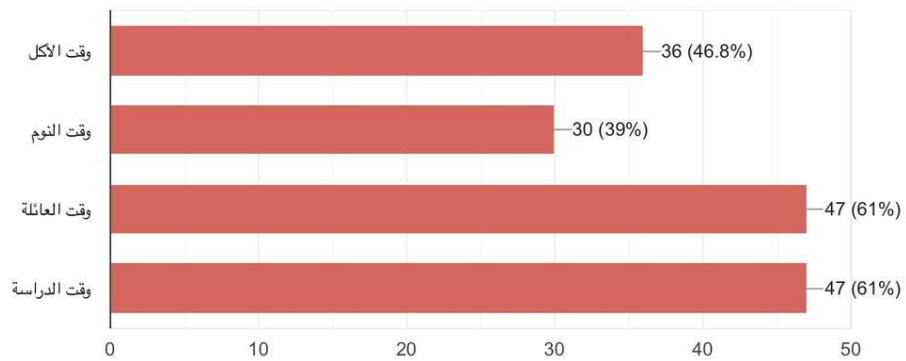


Figure 5 : Results of question 5

٦- ماهي التحديات التي تواجهك في تنظيم وقت الشاشة؟

77 responses

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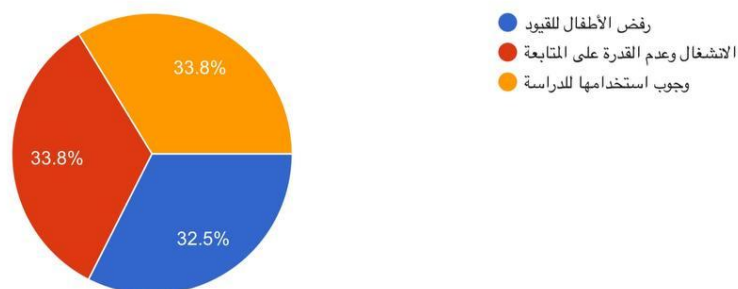


Figure 6 : Results of question 6

٧- ما الميزات التي تجدها ضرورية في التطبيق لتنظيم وقت الشاشة؟

77 responses

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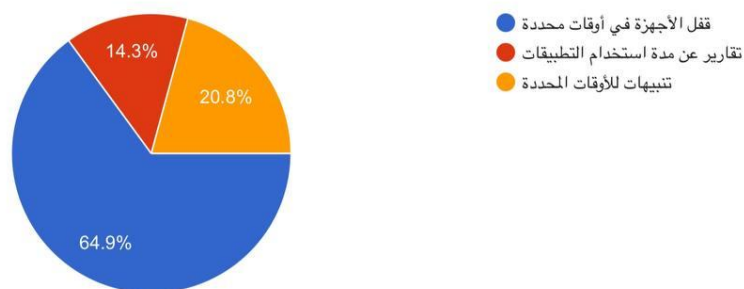


Figure 7 : Results of question 7

٨ - هل ترى أن وجود ميزة تربط وقت الشاشة بمهام معينة (مثل إنهاء الواجبات) سيكون مفيداً؟

77 responses

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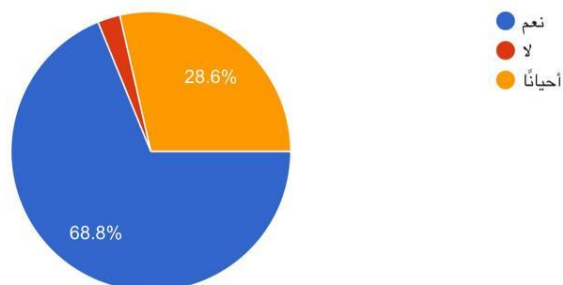


Figure 8 : Results of question 8

4. Personas

These personas highlight the real struggles of modern parents in managing their children's screen time while juggling busy careers and family responsibilities.

- a) **Leena (27, Marketing CEO, Riyadh, mother of two):** She feels overwhelmed by her demanding job, which leaves her with little time to monitor her kids' online activities. She wants to create a structured routine that balances screen time with offline activities like reading, family time, and outdoor play.
Leena's persona was shaped by Abdulrahman's (50) interview, highlighting his frustration with his children's excessive screen time on YouTube and Instagram (Q2a, Q3). He seeks better tools to manage content and enforce screen time limits (Q5, Q6), and is concerned about the digital content's impact on their behavior (Q4). He also wants to improve family communication and reduce screen time (Q7). His main pain points include ineffective existing apps (Q8, Q9)

- b) **Ahmad (47, Engineer, Jeddah, father of four):** He worries that excessive screen time is negatively impacting his children's academics and family communication. He struggles to set screen time rules without resistance and wants a solution to help him track their digital habits while promoting a healthier balance.
Ahmed's persona was developed based on the interview with Nojoud (45). From her responses, we noticed a strong concern about her daughter's late-night TikTok usage (Q2a, Q3), which led us to highlight digital content as a main influence. She also emphasized the importance of family time and sleep, which helped us define her values and goals (Q4). Her struggle with enforcing rules (Q5) and the emotional changes she observed in her children (Q6) reflected her frustration and need for better control tools. We also used her suggestions for app features (Q7, Q8) and her comments on sleep disruption (Q9) to shape her pain points and needs. This all helped us form a persona who is involved, concerned, but lacks the technical means to manage her children's screen time effectively.

Ahmed



"Children nowadays feel like screens are part of their personalities."

Motivations

- Managing children screen time easily.
- Reporting children digital activities effectively.
- Enhancing family connection by reducing the children attachment to screens.

Frustrations

- Struggles with setting rules due to them resisting it.
- Worrying about the type of content his children are exposed to online.
- Concerned about the bad effect of screens on their academic level.

Background

Ahmad is a busy engineer as well as being a parent of 4 kids, he struggles in balancing between work and monitoring his kids screen time. Ahmad has some concerns regarding spending more quality time with his family, he feel that they are on the screens 24/7 which badly effects their communication and other important aspects of their lives.

Age : **47**
Gender : **Male**
Occupation : **Engineer**
Location: **Jeddah, KSA**
Marital Status : **Married**
Family : **Wife , 4 kids**

Needs

A simple, effective app that allows scheduling screen time and locking devices automatically during family time, bedtime, and meals.
Reports that show which apps his children are using and how much time they spend on each.
Features to monitor content exposure

Goals

Ahmad wants to create a better balance between screen time and daily activities for his kids. He hopes to set clear limits without arguments and find an easy way to track their online activities. He also wants to make sure they see safe content and spend more quality time as a family.

Figure 9 : Ahmed's persona

Leena



*"Life is what happens when
you're busy making other
plans."*

Age: **27**

Gender: **Female**

Occupation : **Marketing CEO**

Location: **Riyadh, KSA**

Marital Status: **Married**

Family: **Husband , 2 kids**

Motivations

- Get to be more aware of what society holds .
- To know how to keep an eye on kids .
- Feel reassured that her kids' habits are under control
- Gain convenient access to the app

Frustrations

- Facing challenges in enforcing screen time limits consistently
- guilt for not being able to manage her children's screen time
- Feeling overwhelmed by the lack of balance

Needs

- A flexible app that allows her to set screen time limits easily and adjust schedules as needed.
- Tools to automatically lock devices during important times like family meals, bedtime, and prayer.
- Insights into her children's screen habits without being overly intrusive.

Background

Leena, a 27-year-old mother and MIS graduate from the College of Business at KSU, is a busy CEO with significant responsibilities. Balancing her demanding career leaves her with little time to monitor what her children watch or how they spend most of their time, even though she wishes to stay informed and ensure their safety

Goals

Leena wants to help her kids use their devices in a healthy way while still making time for sleep, school, and family. She hopes to create a routine where they can enjoy screens without letting them take over their daily lives. Instead of just setting strict rules, she wants to encourage more offline activities like playing outside, reading, or spending time together as a family

Figure 10 : Leena's persona

5. Solution

To address parents' concerns, Ra'i offers a smart, automated solution for managing children's screen time effectively. The app enables scheduled device locking during critical times like bedtime and study hours, reducing excessive usage. With real-time app usage reports, parents gain clear insights into their child's digital habits. To ease resistance, Ra'i includes gradual shutdown notifications, allowing children to prepare before screen time ends. Additionally, the app introduces a task-based reward system, encouraging children to complete homework or chores in exchange for screen time. By combining flexibility, automation, and parental control, Ra'i creates a stress-free approach to maintaining a balanced digital lifestyle for families.

6. User Stories

a) User Story 1

As a father, I want to automatically lock my children's devices during family time and bedtime, so I can ensure they have quality time with us and enough rest at night.

b) User Story 2

As a mother concerned about my children's digital habits, I want to track their screen time, so I can guide them towards healthier usage.

c) User Story 3

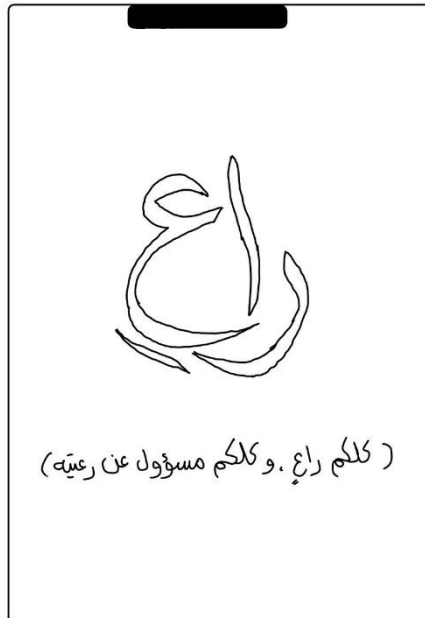
As a father who worries about online safety, I want to keep an eye on the content they are exposed to, so I ensure their safety from unappropriated content online.

d) User Story 4

As a working mother, I want to set clear schedules for my children screen time, so I can avoid everyday arguments with them.

7. Sketches

Figure 11 : Sketch 1 : Start screen



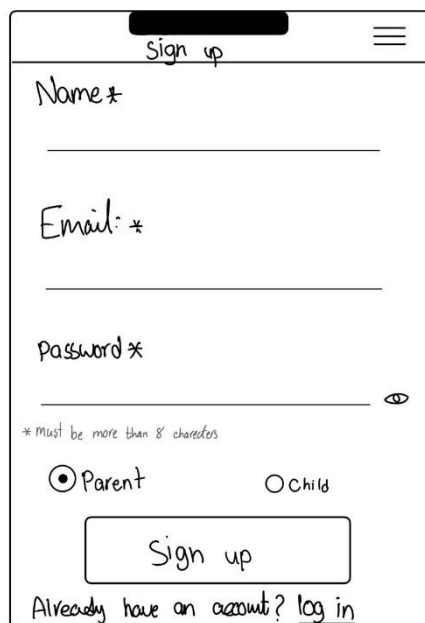
1

Figure 12 : Sketch 2 : Log in screen



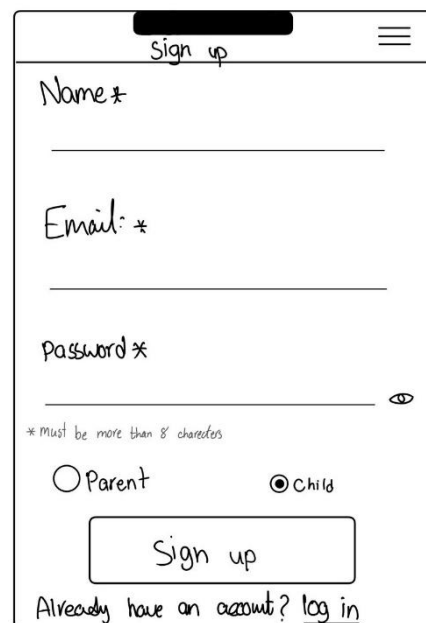
2

Figure 13 : Sketch 3 : sign up screen for parent



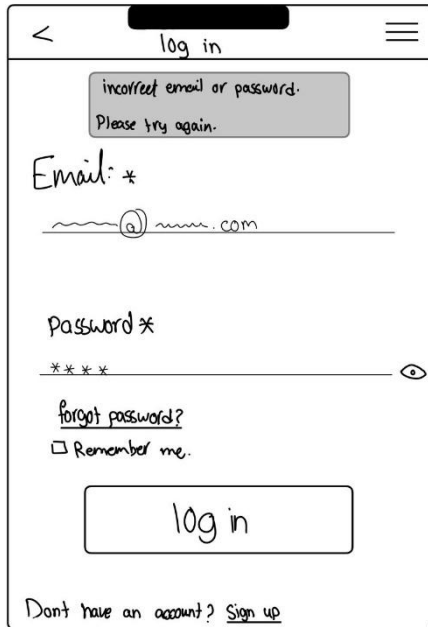
3

Figure14: Sketch 4 : sign up screen for child



4

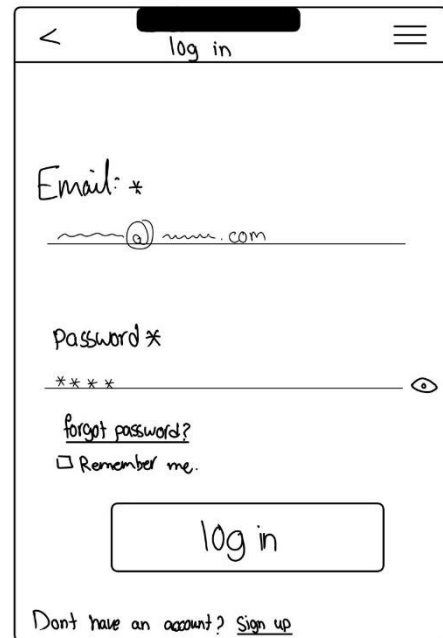
Figure 15 : Sketch 5 : Wrong insert screen



A hand-drawn sketch of a mobile app login screen. At the top, there's a header bar with a back arrow, the text "log in", and a menu icon. Below the header, a grey message box contains the text "incorrect email or password. Please try again.". The main form has two input fields: "Email: *" with a placeholder "~~~~~@~~~~~.com" and "password *" with a placeholder "****" and an eye icon. Below the password field, there's a link "forgot password?" and a checkbox "Remember me.". At the bottom of the form is a "log in" button. At the very bottom, there's a link "Don't have an account? Sign up".

5

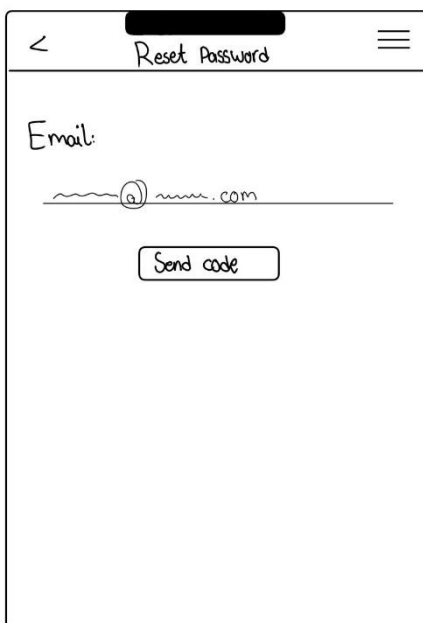
Figure 16 : Sketch 6 : Successful login



A hand-drawn sketch of a mobile app login screen, identical to Sketch 5. It features a header bar with a back arrow, the text "log in", and a menu icon. A grey message box at the top says "incorrect email or password. Please try again.". The form includes "Email: *" and "password *" fields with placeholders and an eye icon, a "forgot password?" link, a "Remember me." checkbox, a "log in" button, and a "Don't have an account? Sign up" link at the bottom.

6

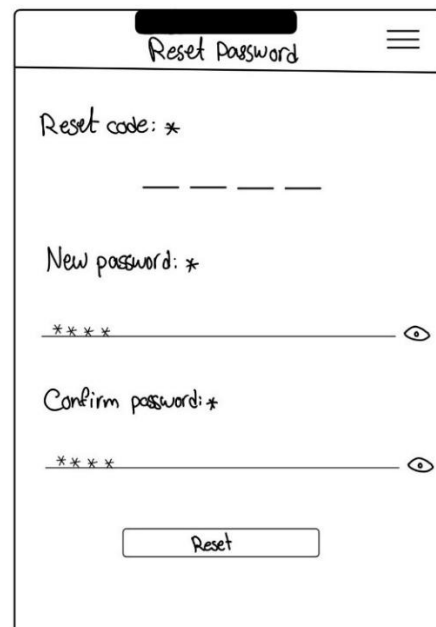
Figure 17 : Sketch 7 : Forgetting password



A hand-drawn sketch of a mobile app "Reset Password" screen. The header bar has a back arrow, the text "Reset Password", and a menu icon. The form has an "Email:" field with a placeholder "~~~~~@~~~~~.com" and a "Send code" button. Below the button, there's a large empty space for a message or code.

7

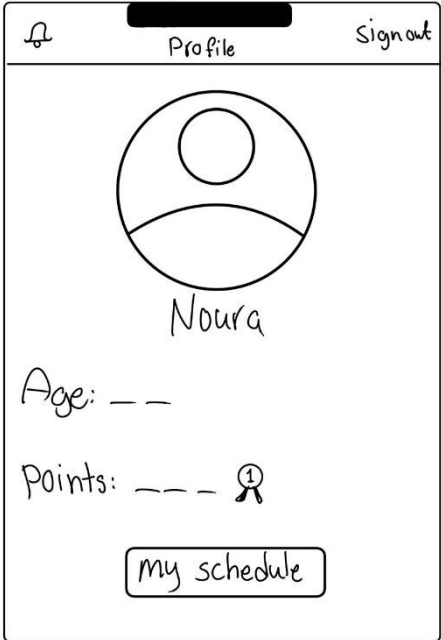
Figure 18 : Sketch 8 : Resetting password



A hand-drawn sketch of a mobile app "Reset Password" screen. The header bar has a back arrow, the text "Reset Password", and a menu icon. The form has three input fields: "Reset code: *" with a placeholder "_____", "New password: *" with a placeholder "****" and an eye icon, and "Confirm password: *" with a placeholder "****" and an eye icon. At the bottom is a "Reset" button.

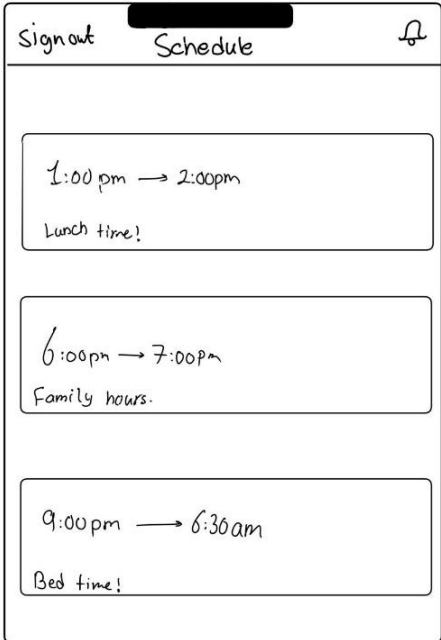
8

Figure 19 : Sketch 9 : Child's Home page



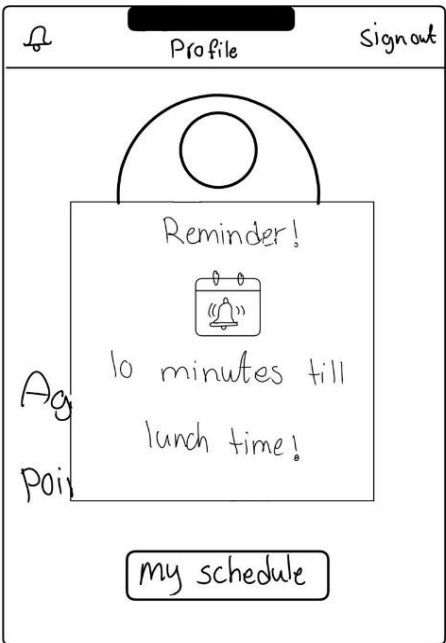
9

Figure 20 : Sketch 10 : Child's schedule page



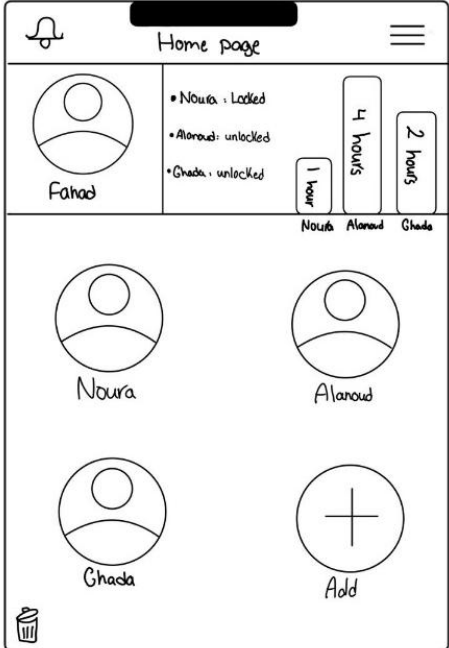
10

Figure 21 : Sketch 11 : Child's reminder screen



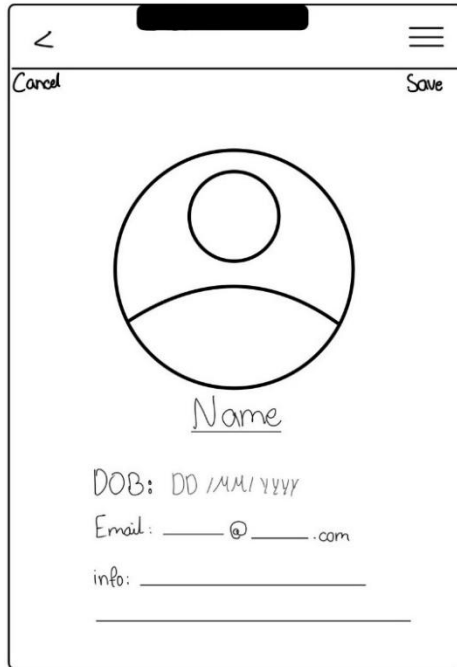
11

Figure 22 : Sketch 12 : Parents' Home screen



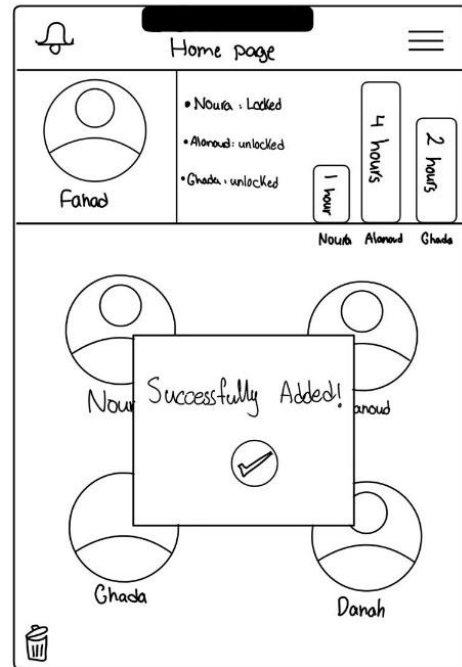
12

Figure 23 : Sketch 13 : Adding a child screen



13

Figure 24 : Sketch 14 : Successful adding notification



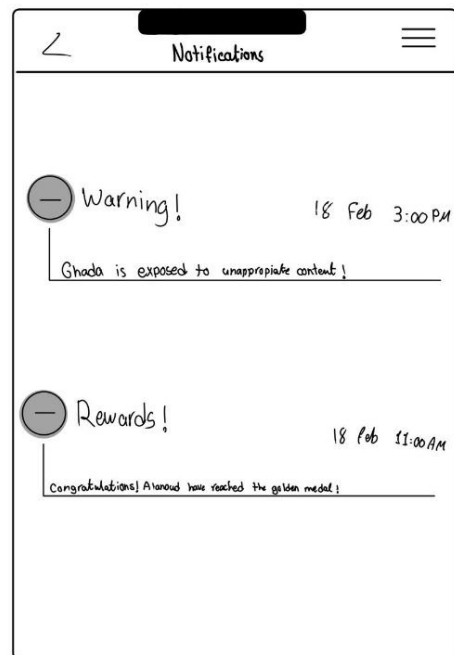
14

Figure 11 : Sketch 15: Menu screen



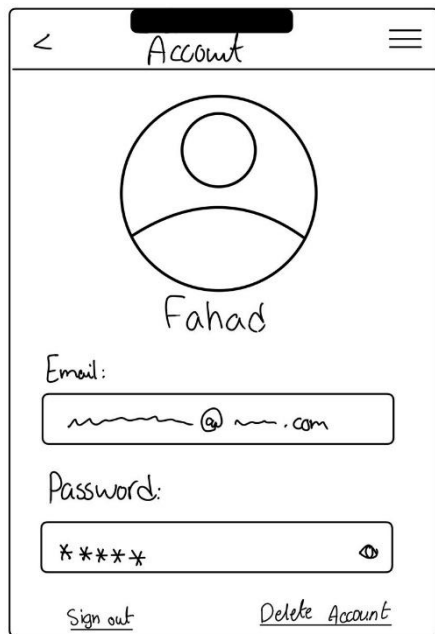
15

Figure 12: Sketch 16 : Notifications screen



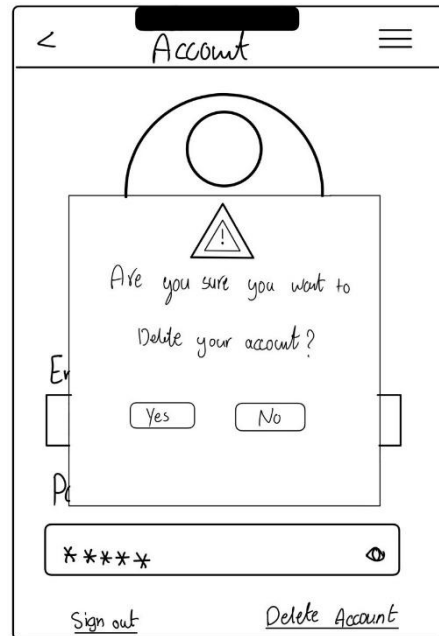
16

Figure 27: Sketch 17 : Parents account page



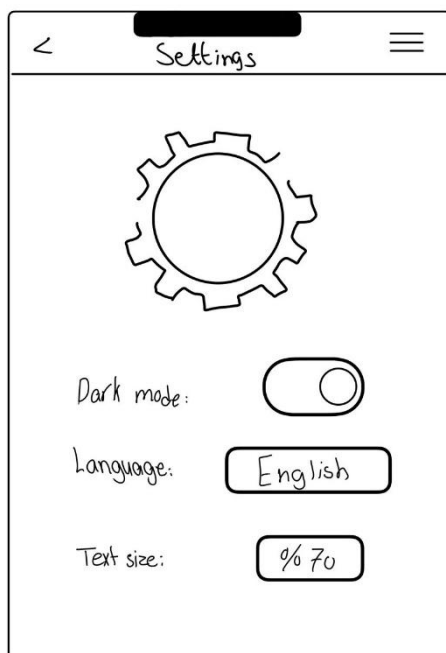
17

Figure 28: Sketch 18 : Deleting account confirmation



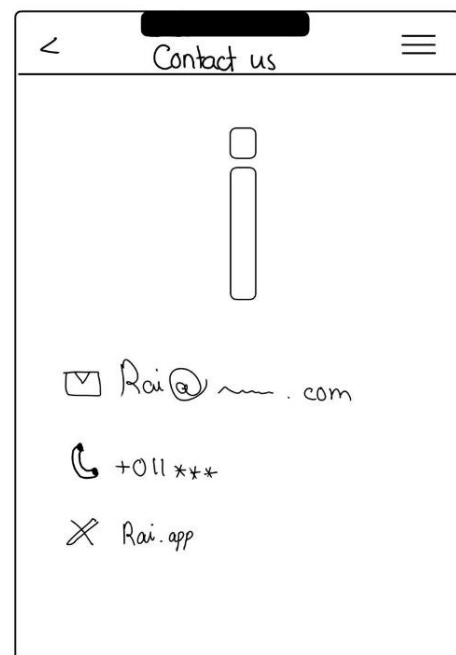
18

Figure 29 : Sketch 19 : Settings page



19

Figure 30 : Sketch 20 : Contact us page



20

Figure 31 : Sketch 21 : Specific child's page



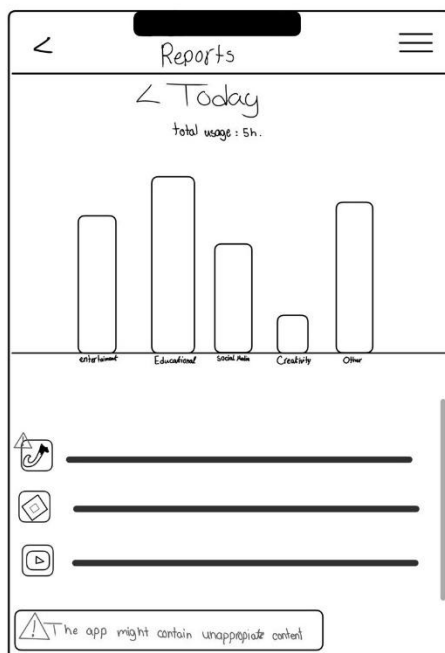
21

Figure 32 : Sketch 22 : Locking the child's device



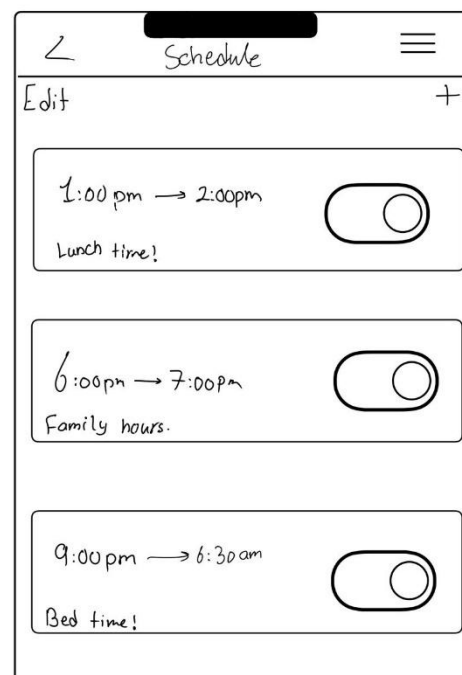
22

Figure 33 : Sketch 23 : Child's reports page



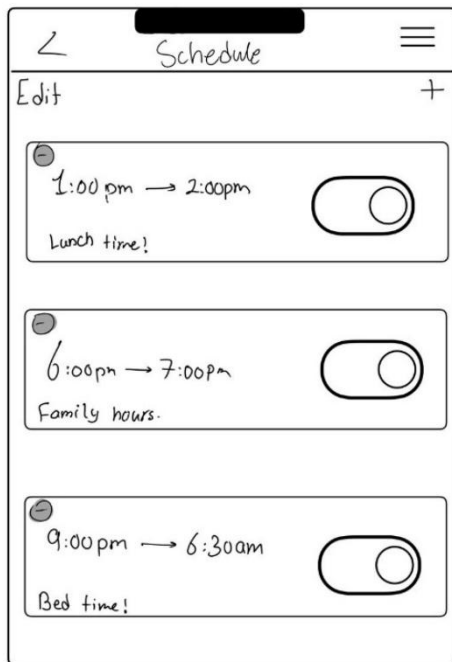
23

Figure 34 : Sketch 24 : Scheduling page



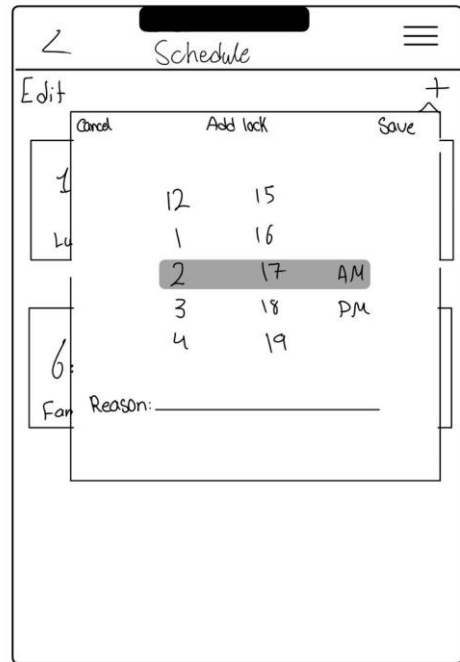
24

Figure 35 : Sketch 25 : Editing the schedule



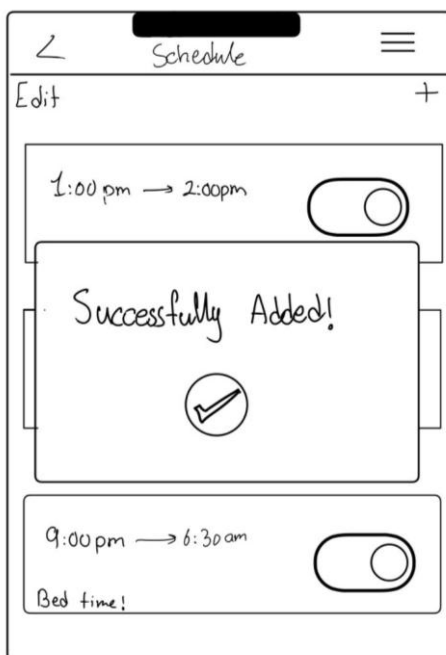
25

Figure 36 : Sketch 26 : Adding a schedule time



26

Figure 37 : Sketch 27 : Successful adding a schedule notification

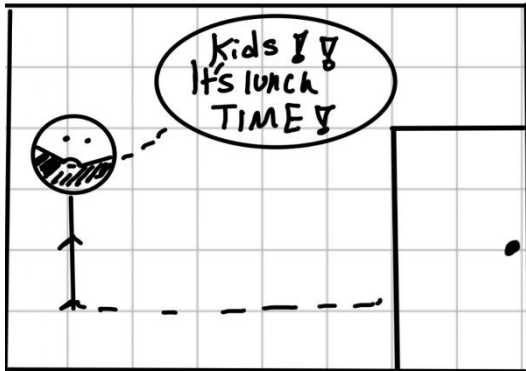


27

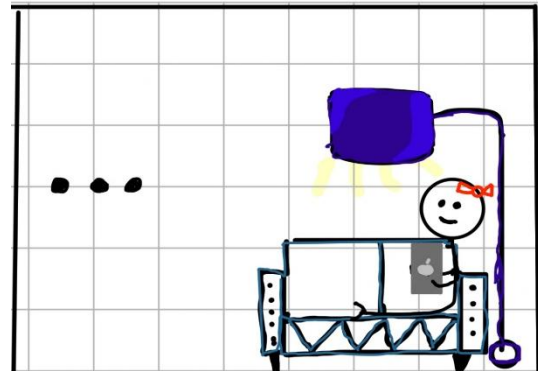
8. Story Board

a) Story Board #1

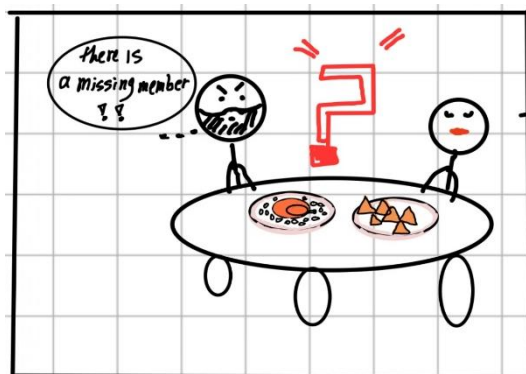
As a father, I want to automatically lock my children's devices during family time and bedtime, so I can ensure they have quality time with us and enough rest at night.



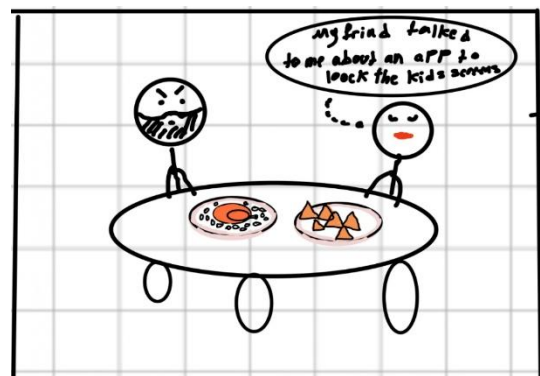
1. Ahmed calls his kids to lunch



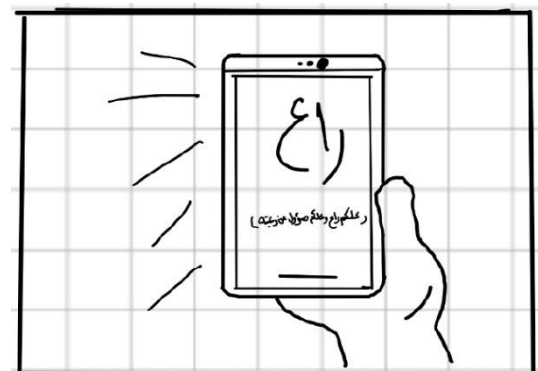
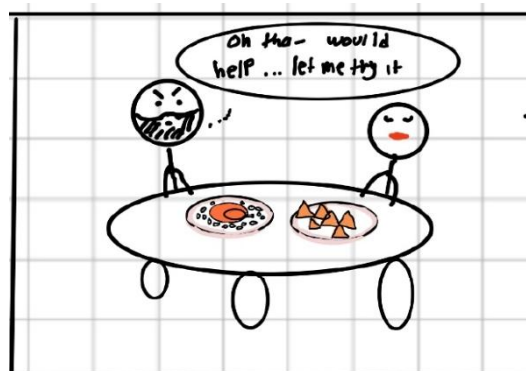
2. They are so absorbed in their devices, as if they are living in their own world



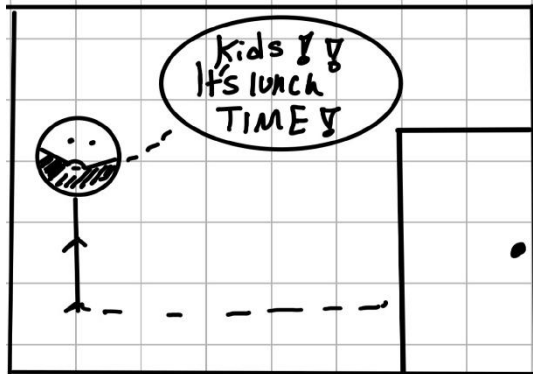
3. Ahmed gets frustrated every time because the family can no longer gather in lunch tables like they used to



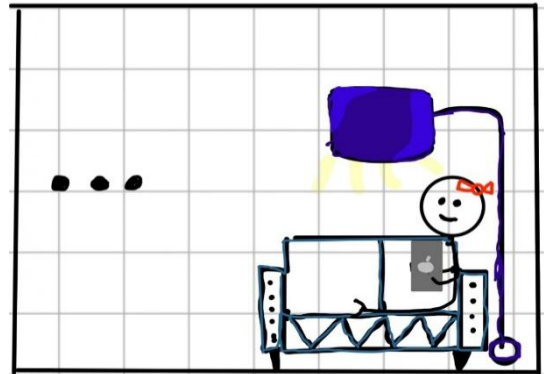
4. Ahmeds' wife suggested using an app to control their children's screen time after hearing about it from a friend



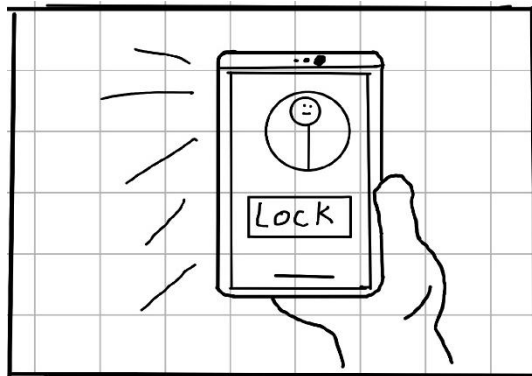
5. Ahmed was curious he had to give the app a chance



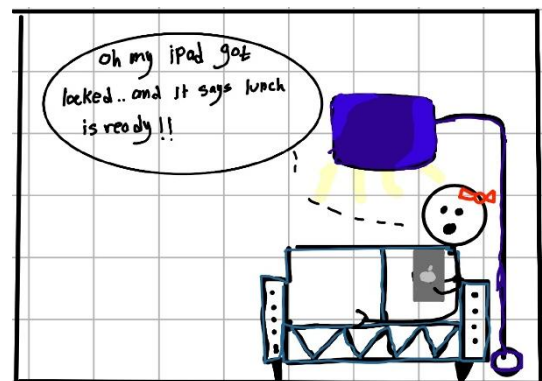
6. Ahmed downloaded and opened the app



7. A few days later, Ahmed called his kids for lunch again

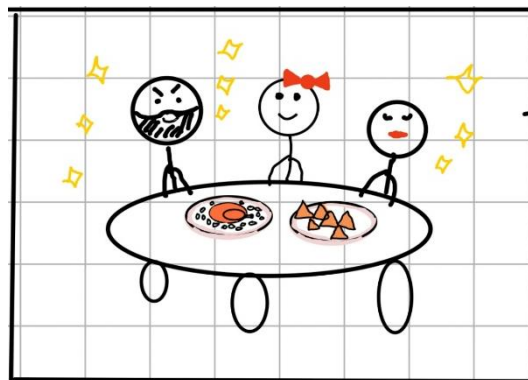


8. The child didn't respond, still busy with the games as usual



9. The device got locked, forcing the child to stop

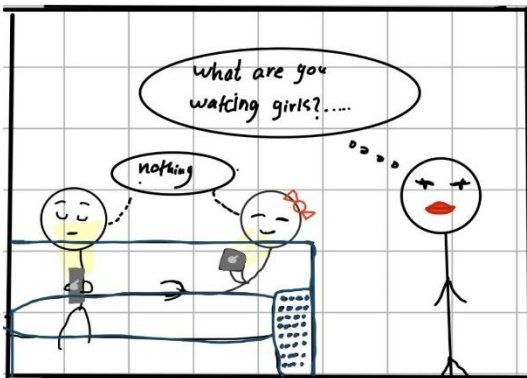
10. The child's digital bubble finally broke, and she realized it was time for lunch



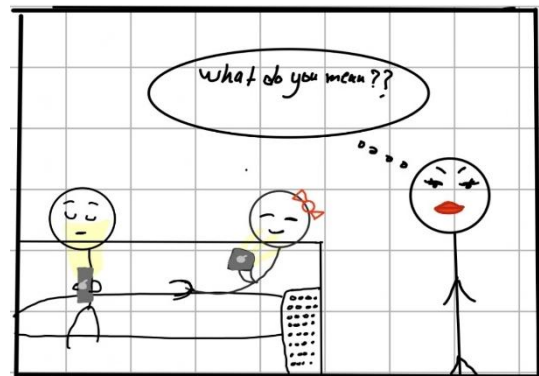
11. The family is finally gathered at the dining table, enjoying their meal without distraction. They can now see each other, talk, and share moments as family

b) Story Board #2

As a mother concerned about my children's digital habits, I want to track their screen time, so I can guide them towards healthier usage.



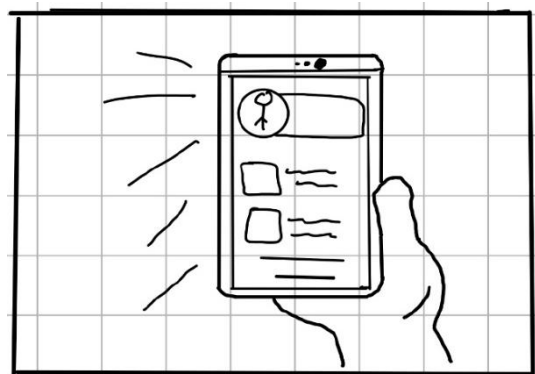
1. The mother is always concerned about the websites and content her daughters access on their devices



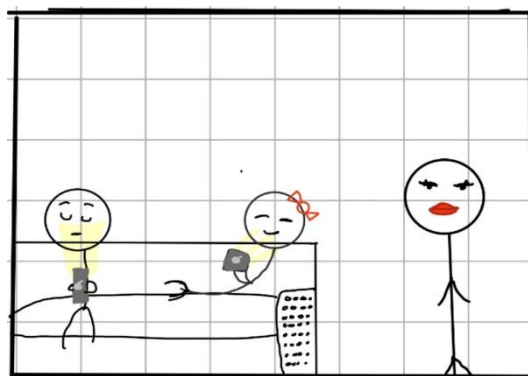
2. She frequently asked them what they were watching, but they seem annoyed by her constant questioning



3. To ease her worries she decided to use Ra'i



4. It provides reports on the content they spend time on and alerts her if they visit any suspicious website



9. User Flow

5. Mom feels relieved, knowing she can monitor their online activity without bothering them, giving her peace of mind

A. User Story #1

As a father, I want to automatically lock my children's devices during family time and bedtime, so I can ensure they have quality time with us and enough rest at night.

A1

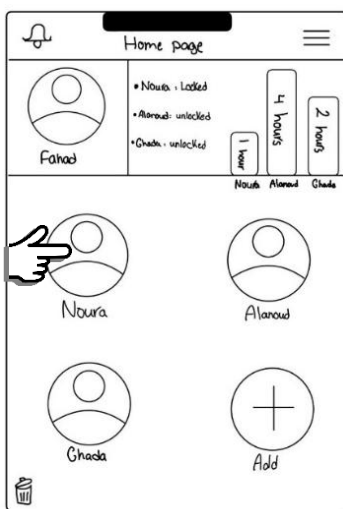


Figure 22 : Sketch 12 : Parents' Home screen

A2



Figure 31 : Sketch 21 : Specific child's page

A3

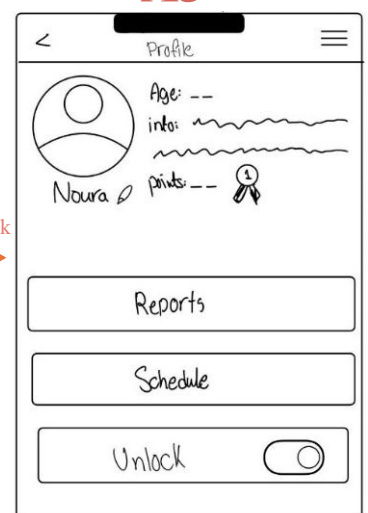


Figure 32 : Sketch 22 : Locking the child's device

Pressing Lock button

Scheduled lock

A6

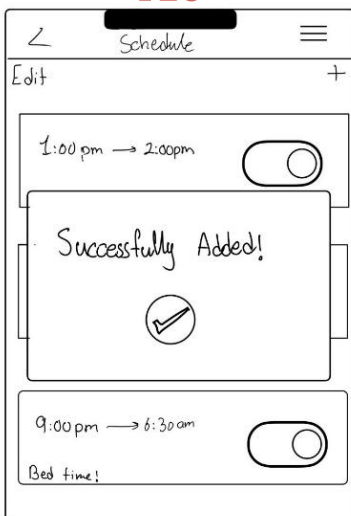


Figure 37 : Sketch 27 : Successful adding a schedule notification

A5

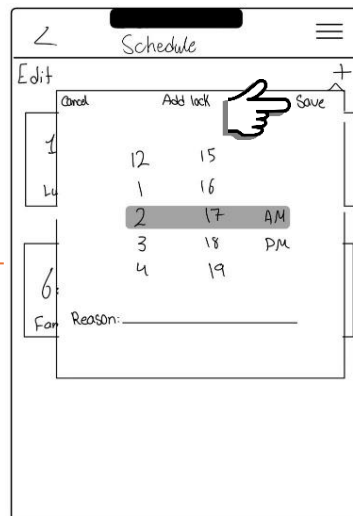


Figure 36 : Sketch 26 : Adding a schedule time

A4

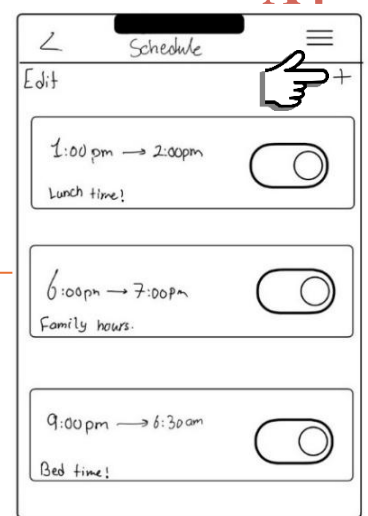
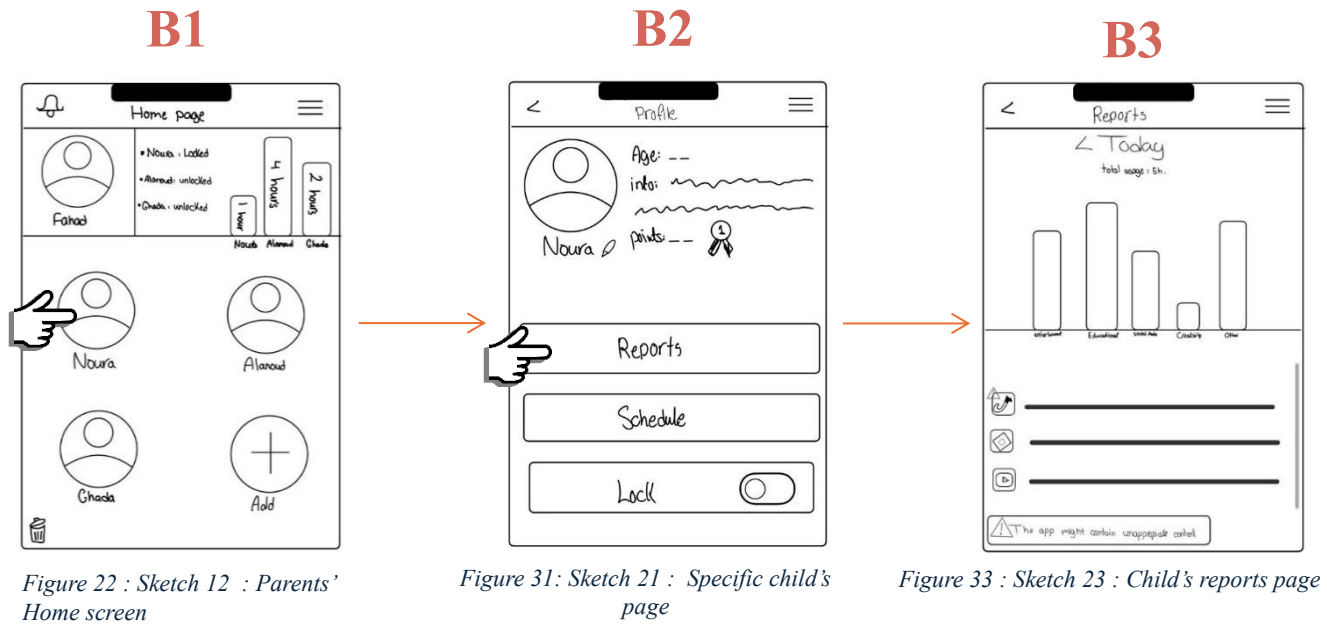


Figure 34 : Sketch 24 : Scheduling page

B. User Story #2

As a mother concerned about my children's digital habits, I want to track their screen time, so I can guide them towards healthier usage.



C. User Story #3

As a father who worries about online safety, I want to keep an eye on the content they are exposed to, so I ensure their safety from unappropriated content online.

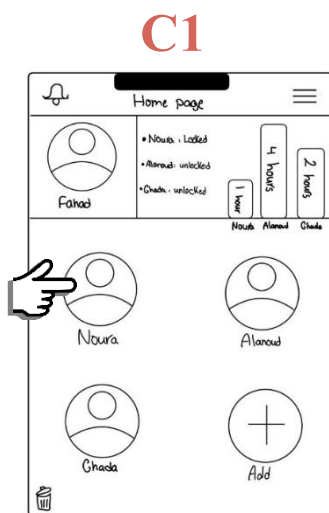


Figure 22 : Sketch 12 : Parents' Home screen



Figure 31 : Sketch 21 : Specific child's page

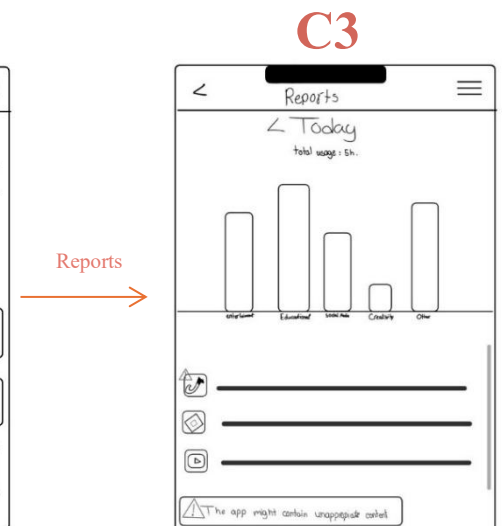


Figure 33 : Sketch 23 : Child's reports page

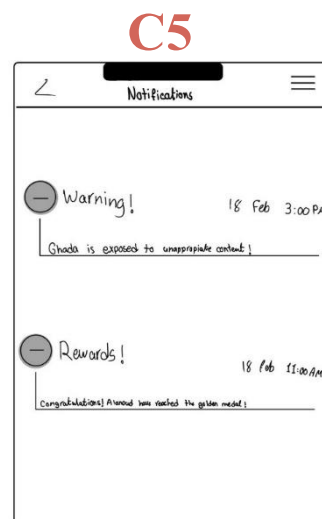
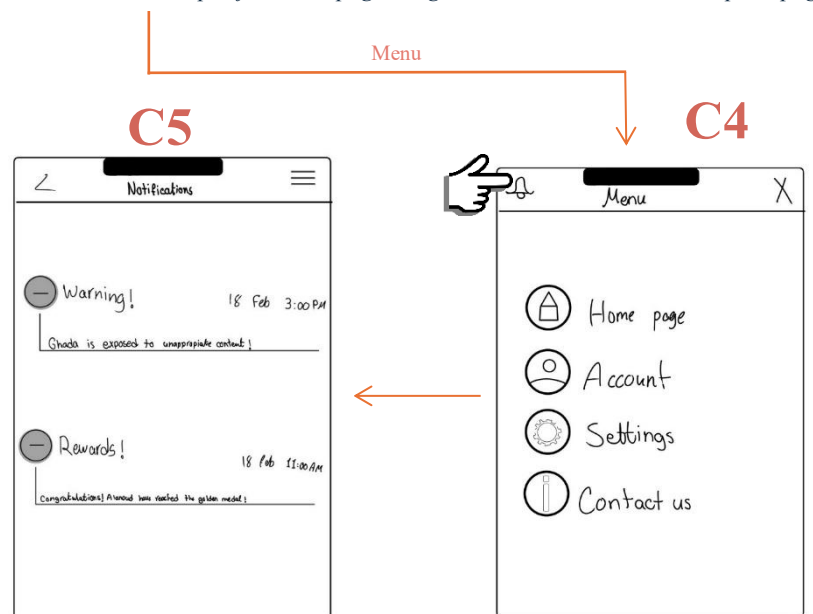


Figure 26: Sketch 16 : Notifications screen



Figure 25 : Sketch 15: Menu screen

D. User Story #4

As a working mother, I want to set clear schedules for my children screen time, so I can avoid everyday arguments with them.

Parent View

D1

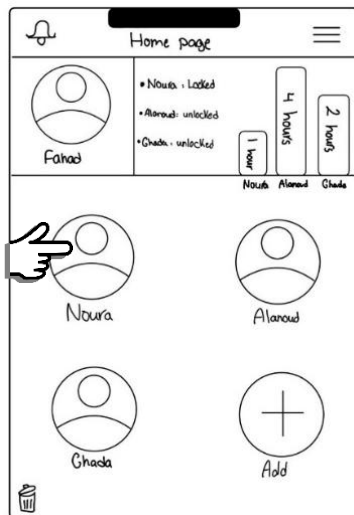


Figure 22 : Sketch 12 : Parents' Home screen

D2

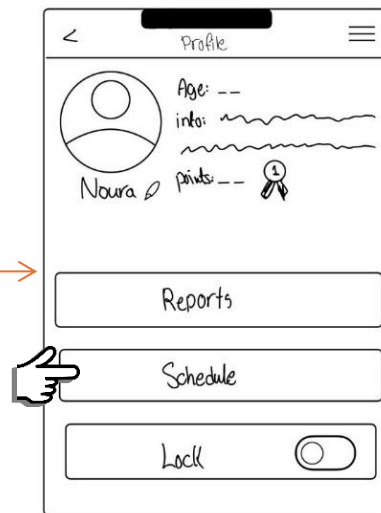


Figure 31 : Sketch 21 : Specific child's page

D3

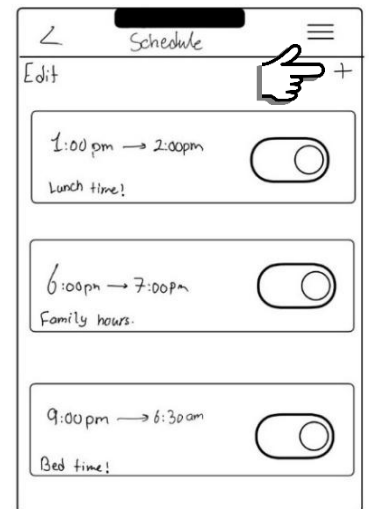


Figure 34 : Sketch 24 : Scheduling page

D5

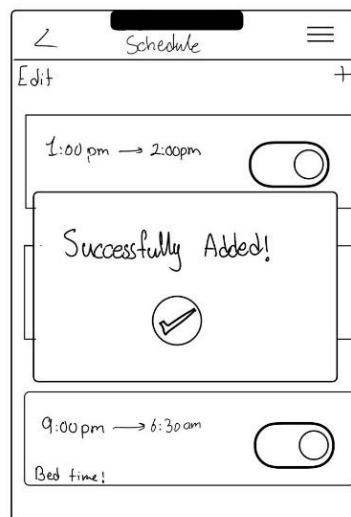


Figure 37 : Sketch 27 : Successful adding a schedule

D4

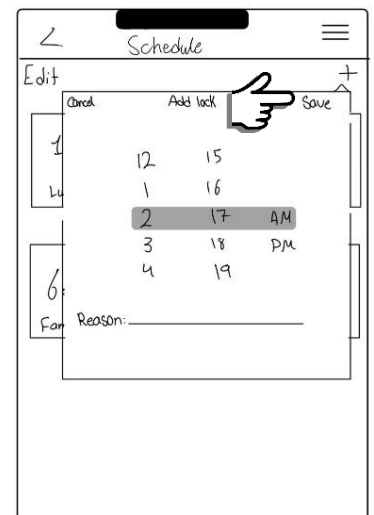


Figure 36 : Sketch 26 : Adding a schedule time

Child View

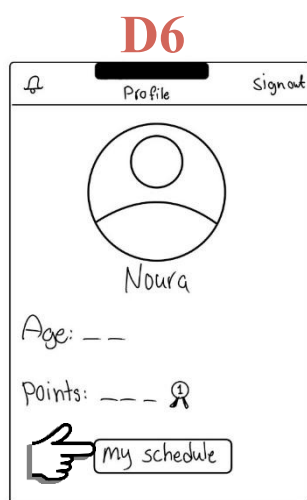


Figure 19 : Sketch 9 : Child's Home page

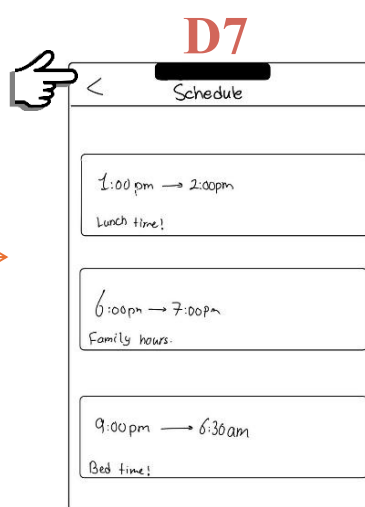


Figure 20 : Sketch 10 : Child's schedule page

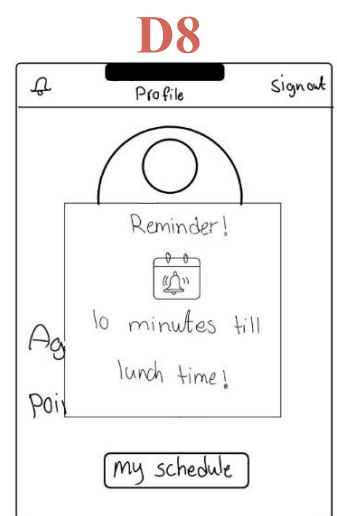


Figure 21 : Sketch 11 : Child's reminder screen

10. Design Rules

Table 1 : Learnability Principles

Learnability Principles

Rule#	Principle		Sketch No/ User Flow No	Describe how you applied this principle
1 2	Predictability		Sketch No.21,22 Flow No.A2,A3 Sketch No.26 Flow No.A5,D4	1. The lock/ unlock feature is designed using a clear toggle switch, making it easy for parents to predict its function. 2. The schedule creation screen follows a common calendar layout, allowing users to intuitively add time slots for device access.
		Operation visibility	Sketch No.3,4,5,6 Sketch No.21,22 Flow No.A2,A3	1. When signing up/logging in the conform button is not activated until all the required information are applied. 2. The lock/unlock status is instantly visible through color-coded indicators.
3	Synthesizability		Sketch No.12,13,14 Sketch No.10 Flow No.D7	1. When a parent adds a new child an immediate notification pops, indicating a successful function. 2. When a parent adds a new restriction or

			schedule, the changes eventually reflect in the child's device list.
4	Familiarity	Sketch No.12 Sketch No.15,16 Flow No.C4,C5	1. Common icons for deletion, such as the dust bin icon for removing a child, it makes navigation intuitive. 2. The bell icon resembles the notification section in most of the apps, reducing the learning curve.
5	Generalizability	Sketch No.21,22,24 Sketch No.24.25.26 Flow No.A4,D3	1. Parents who used the toggle switch in locking/unlocking will extend their knowledge using this switch in other situations in the app such as enabling schedules. 2. The schedule page provides add/edit functions in general representation reflecting variety of apps across the world.
6	Consistency	All Sketch Except No.9,10,11,15 All Flow Except No,C4,D6,D7,D8 Sketch No.15 Flow No.C2,C4	1. Every screen follows a uniform structure, with the same Menu button placements at the top right 2. The action flow follows the icon placement

			ensuring smooth transitions.
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Table 2 : Flexibility Principles

Flexibility Principles

Rule#	Principle	Sketch No/ User Flow No	Describe how you applied this principle
7	Dialog initiative	Sketch No.3,4,5,6,7,8	When the user first sign up, the system manages a conversation leading to less freedom to the user and less errors while signing up due to the importance of the operation.
8	Multi-threading (<i>if applicable</i>)	Sketch No.12	Parents can monitor multiple children's devices simultaneously, switching between profiles in the same session.
9	Task migratability (<i>if applicable</i>)	Sketch No.24,25,26,27 FlowNo. A4,A5,A6 D3,D4,D5	Parents can add a schedule that automatically locks children's screens at certain times, leaving the routine tasks to the system.
10	Substitutivity	Sketch No.21,22 24,25,26,27 Flow No.A2,A3 A4,A5,A6 D3,D4,D5	The app allows parents to choose different ways to manage device control, such as using a direct lock button or pre-setting scheduled restrictions.
11	Customizability	Sketch No.24,25,26,27 Flow No. A4,A5,A6 D3,D4,D5	Parent can customize access levels per device, allowing different rules for school days vs. weekends.

Table 3 : Robustness Principles

Robustness Principles

Rule#	Principle		Sketch No/ User Flow No	Describe how you applied this principle
12 13 14 15	Observability	Browsability	Sketch No.23 Flow No.C2,C3	The parent dashboard provides a real-time overview of children's devices status (locked, unlocked) and their daily screen time.
		Defaults	Sketch No.21,22,25,27 Flow No.D4,D5,A2,A3	The children's lock status is unlocked by default until direct lock or scheduled lock.
		Reachability	Sketch No.15 Flow No.C2,C4	If a parent gets lost in many previous tabs, the menu bar would make them reach what they desire.
		Persistence	Sketch No.23 Flow No.B3,C3	In children's reports, if a child is exposed to an app that might contain inappropriate content, a warning notification would appear right above the app icon and in parents' notification centre.
16	Recoverability		Sketch No.18	When a parent tries to delete the account, a confirmation pop-up appears to prevent accidental changes.
17	Responsiveness		Sketch No.16 Flow No.C4,C5	Notifications update parents whenever a restriction is enforced,

			ensuring timely communication.
18	Task Conformance	Sketch No.27 Flow No. A6,D5	When adding a scheduled lock, a message pops conforming it's successfully added.

11. Prototype

Our UX design includes the following features:

User Account Management

Users can sign up to create a new account as either a parent or a child, or sign in if they already have one. If they forget their password, they can reset it by selecting Forgot Password.

Parent Account Features

1. Menu Options
 - In the Account section, parents can view and manage their account details, including their name, profile character, and other personal information. They also have the option to sign out or delete their account.
 - The Settings section allows parents to personalize their experience by selecting a preferred language, toggling between dark and light modes, and adjusting text size.
 - The Contact Us section provides Ra'i's contact information.
2. Notification Center
 - Parents can stay updated on their child's latest activities and reports.
3. Adding and Deleting a Child
 - By selecting the Add a Child button, parents can link a new child's device for monitoring.
 - By pressing the junk icon , parents can delete any child's device.
4. Child Profile Management
 - When accessing a specific child's profile, parents can:
 - View their child's screen time reports.
 - Edit and add schedules for device usage.
 - Lock or unlock the child's device as needed.

Child Account Features

1. Notification Center

- The child can check their latest updates in the Notification Center located in the upper left corner.

2. Sign Out Option

- The Sign Out button is available in the upper right corner for the child to log out of their account.

3. Profile Page

- The child's profile page displays their information and includes a My Schedule button, where they can see the schedule set by their parents.

4. Reminders & Notifications

- A Reminder Notification will appear 10 minutes before a scheduled time to notify the child of upcoming restrictions or activities.

We used **'Figma'** and **'previewed'** to build our prototype.

Youtube Video : <https://youtu.be/dQpYxQB4Z7U?si=OWUKnf9nJKo0skPk>

Prototype Design: <https://www.figma.com/proto/pcb7cr0LxDDuzc1jV4mpVT/Prototype?node-id=10-94&p=f&t=WGBCFkSrpCuQAocF-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=3%3A4&show-proto-sidebar=1>

12. Testing

Step 1: Participants

Participant Profiles

We conducted our usability testing with 6 participants ranging in age from 29 to 53 years old. Three of them were females (Nujoud, Shahad, and Muneera), and three were males (Abdulrahman, Nawaf, and Mohammad).

Our participants come from diverse backgrounds and varying levels of experience. Shahad and Nawaf had moderate experience with similar products, while Mohammad had higher experience. On the other hand, Nujoud, Abdulrahman, and Muneera had minimal prior experience with similar products.

All participants reported familiarity with mobile and website interfaces, with most scoring 4 or 5 in usage frequency, except Mohammad who scored lower.

<i>Participant name</i>	<i>Age</i>	<i>Gender</i>	<i>Experience with similar products (1-5)</i>	<i>Use of mobile/website interface (1-5)</i>
1 Nujoud	45	Female	1	4
2 Abdulrahman	50	Male	1	3
3 Shahad	29	Female	3	5
4 Nawaf	32	Male	3	5
5 Mohammad	53	Male	4	2
6 Muneera	45	Female	1	4

Step 2: Procedure

Outline of the procedure

After finalizing the design of the Rai screen time monitoring application, we selected six participants from our research sample to conduct a usability test. In preparation, we created the necessary materials, including a consent form, a questionnaire, a test scenario, and a set of tasks for participants to complete during the session.

We scheduled individual sessions with each participant at a time and date that suited them best. At the beginning of each session, we welcomed the participant and provided them with the consent form to obtain their approval to contribute to this study. Before reading the facilitator script, we instructed them to verbalize their thoughts while interacting with the system, explaining their actions and reasoning as they completed the tasks.

Next, we introduced the scenario and presented the five tasks they needed to complete using the application. Once the participant finished the tasks, we asked them to fill out a questionnaire and conducted a short interview to gather additional insights. Finally, we expressed our gratitude for their time and valuable feedback.

Timeline

The test was conducted between 25th of March to 27th of March 2025, Shahad finished the test in 3.6 minutes which is the least test time, while Mohammed finished the test in 7.5 minutes. Session's time varied between 3.6 – 7.5 minutes, with an average of 5.4 minutes.

Participant name	Day	Date	Start session time	Finish session time	Duration of session
1 Nujoud	Tuesday	25 Mar 2025	8:00PM	8:05PM	5.5 m
2Abdulrahma	Tuesday	25 Mar 2025	8:30PM	8:37PM	7.13 m
3 Shahad	Wednesday	26 Mar 2025	8:00PM	8:03PM	3.6 m
4 Nawaf	Wednesday	26 Mar 2025	8:30PM	8:34PM	4 m
5 Mohammad	Thursday	27 Mar 2025	8:00PM	8:07PM	7.5 m
6 Muneera	Thursday	27 Mar 2025	8:30PM	8:35PM	5.25 m

Tasks

To evaluate the usability of our prototype, we conducted a task-based test with participants. Each participant was asked to complete five main tasks that reflect key features of the application.

Task #	Task (as it was presented to users)	Objective of the task (What was this task designed for measuring?)
1	Create a parent account (Sign Up)	To evaluate how easily users can register and sign up as a parent.
2	Add/Delete a child	To test how intuitive it is to manage child profiles within the app.
3	Edit/Add a scheduled lock	Test how users set screen time limits.
4	View Today and Yesterday Report	To assess how easily users can access detailed reports on the child's device and app usage.
5	Delete your account	Test how users delete their account.

Performance Measures and Feedback

The performance of the system was evaluated using three key metrics: effectiveness, efficiency, and user satisfaction. Effectiveness was measured by tracking the number of successfully completed tasks and counting errors made during each task. Efficiency was assessed by recording the time taken to complete each task and calculating the average completion time. To measure user satisfaction, a survey was conducted to gather participants' feedback on their overall experience.

Metric	How was it measured?	How was the data created/captured?
Effectiveness	Task success	By checking whether participants completed the task or not.
	Number of Errors	Observed during testing and recorded based on user interactions.
Efficiency	Time on Task	Recorded using a stopwatch/timer during task performance.
Satisfaction	User surveys	Feedback collected from post-task and post-test satisfaction google form.

Step 3: Usability Results and Discussion

presenting the findings from the usability test, highlighting key observations, user errors, and feedback. This includes an analysis of task performance, common mistakes encountered, and insights gathered from the satisfaction questionnaire. Additionally, a summary of the collected data will provide an overall assessment of the system's usability, identifying strengths and areas for improvement based on user interactions and feedback.

Results:

1. Task Completion Success and Time on Task

All participants successfully completed all the tasks except for Task 3, which was completed by five out of six participants (83.33%). Interestingly, Task 3 also required the longest average completion time, with participants taking approximately 117.4 seconds. In contrast, Task 2 had a perfect completion rate and took the shortest average time to complete likely due to its simplicity and ease of access. Task 4, which required users to view specific information, also had a high success rate and a relatively short completion time of around 42.5 seconds. Meanwhile, Task 1 ranged from 65 to 136 seconds, averaging at 96 seconds. Task 5 had the lowest average time at 33.5 seconds and was completed successfully by all participants. In summary, average task completion times ranged from 33.5 seconds to 117.4 seconds.

Participant	Task 1		Task 2		Task 3		Task 4		Task 5	
	complete task	Time on Task	complete task	Time on Task	complete task	Time on Task	complete task	Time on Task	complete task	Time on Task
P1	✓	97s	✓	37s	✓	115s	✓	49s	✓	30s
P2	✓	111s	✓	54s	✓	161s	✓	56s	✓	46s
P3	✓	65s	✓	17s	✓	94s	✓	24s	✓	19s
P4	✓	79s	✓	24s	✓	84s	✓	27s	✓	26s
P5	✓	136s	✓	58s	-	-	✓	59s	✓	52s
P6	✓	88s	✓	26s	✓	133s	✓	40s	✓	28s
Completion Rates	6/6 100%		6/6 100%		5/6 83.33%		6/6 100%		6/6 100%	
AVG . Time on Task		96s		36s		117.4s		42.5s		33.5s

2. Errors

Tasks 1 and 4 were completed successfully by all participants without any errors. Task 2 included a non-critical issue where one user struggled to identify the delete icon due to its small size. Task 3 presented more challenges, with several participants taking extra time to locate the scheduled lock feature, and one participant failing to complete it due to confusion with the “From” and “To” time setting flow, making this a critical error. As for Task 5, some users experienced a non-critical error as they searched for the delete account option in the wrong section.

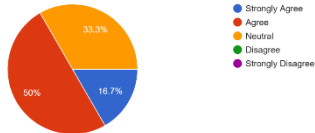
Tasks	Number of errors	Errors description	Critical or non-critical (C,NC)
Task 1	0		
Task 2	1	One user struggled to notice the delete icon as it was relatively small, which caused hesitation during deletion.	NC
Task 3	2	A few users took time to locate the “Set Scheduled” feature.	NC
		One user couldn’t complete the task as the flow of the FROM and TO clock function wasn’t clear enough.	C
Task 4	0		
Task 5	1	Some users took longer to find the “Delete Account” option, they were looking for it in settings instead of account.	NC

3. Satisfaction questionnaire

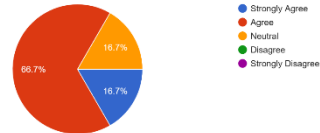
Figure 38 : Post Task 1

Post Task 1:

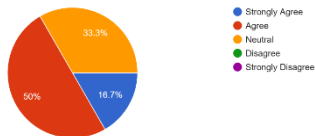
1- The task was easy to complete.
6 responses



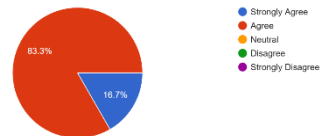
2- I didn't feel confused or lost while performing this task.
6 responses



3- The task flow was smooth and intuitive to follow.
6 responses



4- The buttons and their functions in this task were clear and easy to understand.
6 responses



5- I'm satisfied with the result I got after completing this task.
6 responses

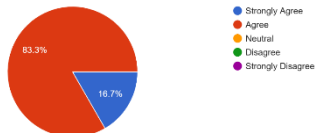
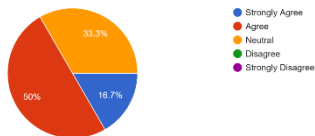


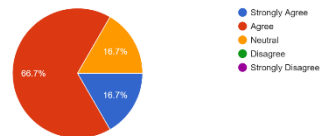
Figure 39 : Post Task 2

Post Task 2:

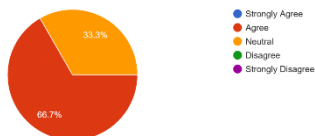
1- The task was easy to complete.
6 responses



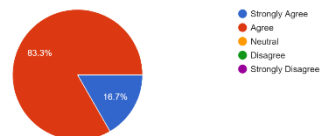
2- I didn't feel confused or lost while performing this task.
6 responses



3- The task flow was smooth and intuitive to follow.
6 responses



4- The buttons and their functions in this task were clear and easy to understand.
6 responses



5- I'm satisfied with the result I got after completing this task.
6 responses

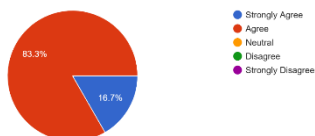
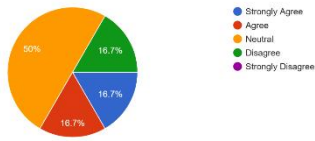


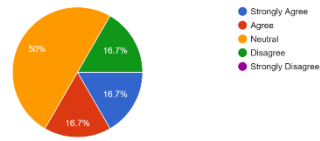
Figure 40 : Post Task 3

Post Task 3:

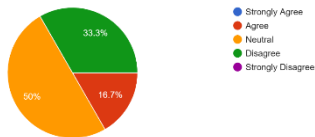
1- The task was easy to complete.
6 responses



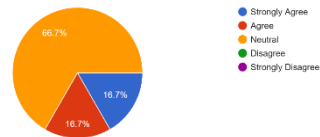
2- I didn't feel confused or lost while performing this task.
6 responses



3- The task flow was smooth and intuitive to follow.
6 responses



4- The buttons and their functions in this task were clear and easy to understand.
6 responses



5- I'm satisfied with the result I got after completing this task.
6 responses

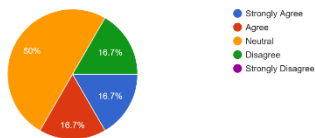
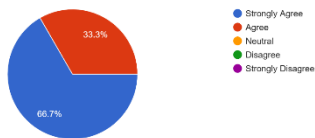


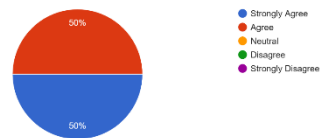
Figure 41 : Post Task 4

Post Task 4:

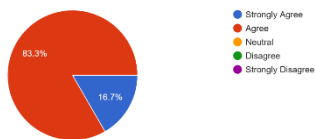
1- The task was easy to complete.
6 responses



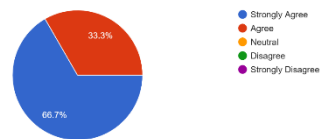
2- I didn't feel confused or lost while performing this task.
6 responses



3- The task flow was smooth and intuitive to follow.
6 responses



4- The buttons and their functions in this task were clear and easy to understand.
6 responses



5- I'm satisfied with the result I got after completing this task.
6 responses

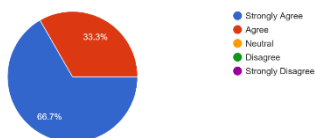
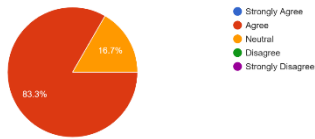


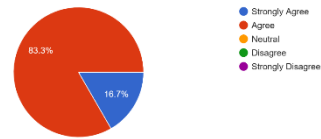
Figure 42 : Post Task 5

Post Task 5:

1- The task was easy to complete.
6 responses



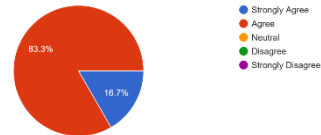
2- I didn't feel confused or lost while performing this task.
6 responses



3- The task flow was smooth and intuitive to follow.
6 responses



4- The buttons and their functions in this task were clear and easy to understand.
6 responses



5- I'm satisfied with the result I got after completing this task.
6 responses

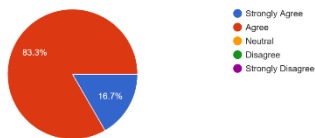
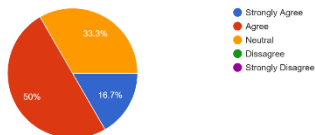


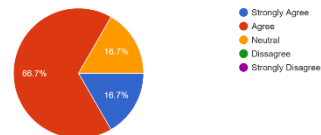
Figure 43 : Post Test

Post Test:

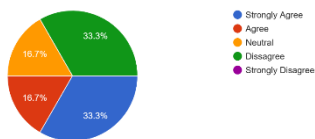
1- The test was easy to complete.
6 responses



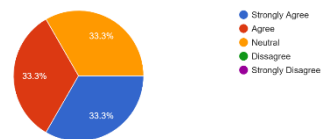
2- I clearly understood what I was supposed to do in this test.
6 responses



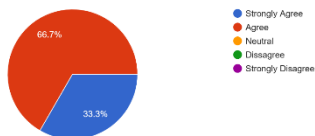
3- The layout and design made it easy for me to complete tasks efficiently.
6 responses



4- The system's interface felt familiar and intuitive to use.
6 responses



5- I would be comfortable using this system again in the future
6 responses



Satisfaction Table:

We asked our participants different questions to measure their satisfaction with each task. All of our tasks got a high satisfaction rate based on user feedback, with Task 4 scoring the highest. Task 3 scored the lowest overall, but still maintained a decent rating.

Notes:

1. The average is calculated out of 5 points.
2. For Q1–Q5, higher scores = better satisfaction
 - “Strongly Agree” = 5 pts, down to “Strongly Disagree” = 0 pts

Task	Question 1 Avg.	Question 2 Avg.	Question 3 Avg.	Question 4 Avg.	Question 5 Avg.	Overall
1	3.83	4	3.83	4.17	4.17	4
2	3.83	4	3.67	4.17	4.17	3.97
3	3.33	3.33	2.67	3.5	3.5	3.27
4	4.67	4.5	4.17	4.67	4.67	4.53
5	3.83	4.17	4	4.17	4.17	4.07

3. Summary of Data

The table below summarizes the test data, highlighting areas of concern in red. These include **lower task completion rates**, **reduced satisfaction scores**, **higher error occurrences**, and **longer task completion times**.

Summary of Completion, Errors, Time on Task, Average Satisfaction

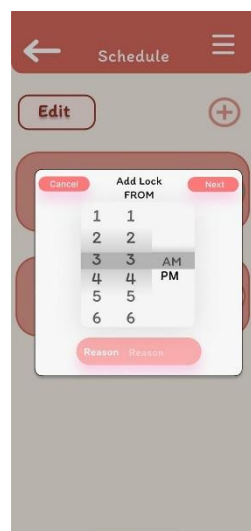
Task	Task Completion	Errors	Time on Task	Satisfaction*
1	6	0	96s	4
2	6	1	36s	3.97
3	5	2	117.4s	3.27
4	6	0	42.5s	4.53
5	6	1	33.5s	4.07

Satisfaction = Average you got from the questionnaire for each task

Discussion:

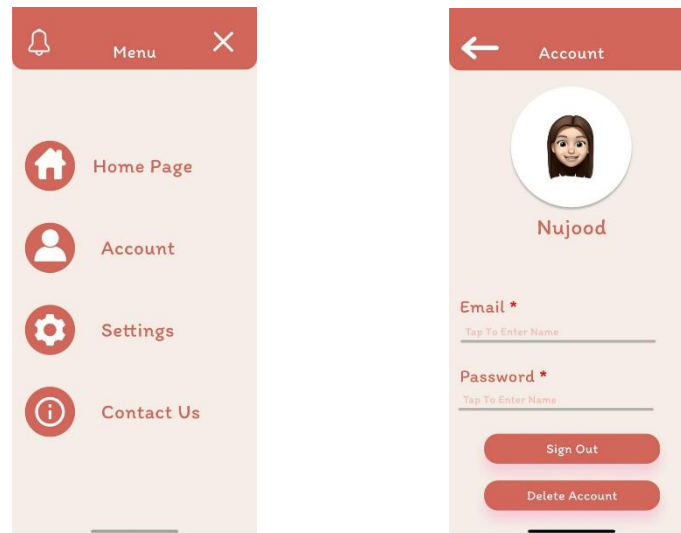
1. A critical problem

Users were confused in task 3 by the split “FROM” and “TO” time selection. Some clicked “Next” without realizing they needed to set both times, which caused incorrect scheduling.



2. A major problem

Most users in task 5 looked for the “Delete Account” option inside Settings instead of the Profile section, which caused delays and confusion while completing the task.



3. A minor problem

In task 2, the delete icon was too small, making it difficult for users to notice, which led to hesitation and uncertainty during deletion.



Step 5: Recommendations

Based on participant feedback, several suggestions were made to improve usability and better align the design with user expectations.

For the critical issue, some users suggested merging the “Start Time” and “End Time” pickers into a single screen, with a brief instruction to clarify the flow. They also recommended disabling the “Save” button until both times are selected.

Regarding the major issue, participants expected the “Delete Account” option to be under Settings. A common suggestion was to move or duplicate the option there to make it easier to find.

For the minor issue, a few users mentioned the delete icon was too small. They suggested increasing its size and visibility to make the action more noticeable and accessible.

Appendix

1- Consent Form:

Goal

The aim of this evaluation is to assess how user-friendly the Ra'i prototype is. This includes observing how easily participants can navigate the app and complete key tasks. The goal is to gather insights that will help improve the user experience and design of the app before moving forward.

Procedure

Participants will complete a set of tasks using the Ra'i prototype while sharing their thoughts out loud.

Afterward, they will fill out a short questionnaire and briefly discuss their experience with the facilitator. The entire session should take about 30 minutes.

Recording

There are no risks involved in participating in this study. Your answers and behavior will be treated confidentially and used only for improving the design of the Ra'i app. We are not judging your abilities the focus is on how the app performs.

With your permission, we may record audio or video during the session, but this is entirely optional.

Informed Consent

By signing this form, you acknowledge that you understand the purpose of this study and agree to take part in it under the terms stated above.

Name:

Do you agree with the recording?

Date:

☐ Yes, video and sound recording

Email:

☐ Yes, only sound recording

Signature:

☐ No, rather not

2- Facilitator Script:

Scenario

Rana is a working mother with two kids who often use their tablets unsupervised. She worries they're spending too much time on entertainment apps and staying up late on school nights. To regain control, Rana installs the Ra'i app, which helps her manage screen time through scheduled locks and gives her access to usage reports.

With the help of Ra'i, Rana sets up screen time limits, checks her children's device activity for the last two days, and makes adjustments to fit the family's routine. She also explores options like account management and child profiles to fully personalize the experience.

Tasks

1. Create a parent account (Sign Up)
2. Add/Delete a child
3. Edit/Add a scheduled lock
4. View Today and Yesterday Report
5. Delete your account

3- Questionnaires Script:

1. The task was easy to complete.

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly Disagree

2. I didn't feel confused or lost while performing this task.

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly Disagree

3. The task flow was smooth and intuitive to follow.

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly Disagree

4. The buttons and their functions in this task were clear and easy to understand.

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly Disagree

5. I'm satisfied with the result I got after completing this task.

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly Disagree

Tools And References

- **Methodology**

- The interviews were conducted by our team interviewing chosen individuals via scheduled one-to-one meetings.
- The questionnaire was conducted using Google Forms and distributed via WhatsApp.

- **Persona**

We used 'Xtensio' tool to conduct the personas.

- **Sketches & Story Board & User Flow**

We used 'CollaNote' tool to conduct the Sketches & Story Board & User Flow.

- **Prototype**

We used 'Figma' and 'previewed' tools to conduct the prototype.

- Keyboards kit: <https://www.figma.com/community/file/1385659531316001292>
- iOS 18 and iPadOS 18 UI kit:
<https://www.figma.com/design/cOvxBWC6CDMobeR1PvWpVL/iOS-Keyboards--Community-?node-id=59-1880&p=f&t=De8Q6lxFO4VCyrtm-0>
- Pinterest Website:
 1. <https://pin.it/4DEsEDJKO>
 2. <https://pin.it/5cQPE8DTc>
 3. <https://pin.it/4IkhEPBcB>
 4. <https://pin.it/4tqFuS6in>

- **Testing**

We used Google Forms tool to conduct the testing satisfaction questionnaire forms and distributed them via WhatsApp.

Appendices:

Appendix A: Interview

1. What do you think about the impact of electronic devices on children's daily lives?
2. What are the main factors influencing how much screen time your children get?
3. Tell me about the last time you felt worried about your children's use of electronic devices. What happened?
4. When do you prefer to limit your children's screen time? And why?
5. Can you describe a situation where you faced a challenge in limiting your children's screen time? How did you handle it?
6. What changes have you noticed in your children's behaviour when they spend too much time on screens? Can you share a specific example?
7. Have you ever used an app to manage screen time? How was your experience?
8. If you had an app to help manage screen time, what feature would be most useful to you? And why?
9. What changes have you noticed in your children's sleep after using electronic devices? Can you share a specific example?
10. Have you ever seen your child try to reduce their screen time on their own? What did they do?

Interviewee 1: Nojoud (45)

Interviewer: Noura Altuwaim

1. What do you think about the impact of electronic devices on children's daily lives?
Devices take up a significant portion of their day, pulling them away from family and schoolwork.
2. What are the main factors influencing how much screen time your children get?
 - a. For my middle daughter (15 years old), it's mainly TikTok, she keeps scrolling endlessly for hours.
 - b. For my youngest, it's video games.
 - c. The older girls spend a lot of time on social media or chatting with friends.
3. Tell me about the last time you felt worried about your children's use of electronic devices. What happened?
My biggest concern is my middle daughter staying up late on TikTok for hours. Sometimes, I don't even know what kind of content she's consuming.
4. When do you prefer to limit your children's screen time? And why?
At bedtime and during family time because sleep is essential for their health, and family time is an opportunity to connect.
5. Can you describe a situation where you faced a challenge in limiting your children's screen time? How did you handle it?

I once tried taking away their devices after 10 PM, but they resisted especially my middle daughter, who was deeply engaged in scrolling through TikTok.

6. What changes have you noticed in your children's behaviour when they spend too much time on screens? Can you share a specific example?

They become more withdrawn and less engaged with us. Sometimes, when I ask them to put their devices away, they get irritated or anxious.

7. Have you ever used an app to manage screen time? How was your experience?

No, but I really need an app that lets me control their screen time and monitor content without being too intrusive.

8. If you had an app to help manage screen time, what feature would be most useful to you? And why?

A feature that limits the time spent on apps, along with reports on what website they visit.

9. What changes have you noticed in your children's sleep after using electronic devices? Can you share a specific example?

Their sleep schedule has been significantly delayed, especially my middle daughter, who stays up late on social media.

10. Have you ever seen your child try to reduce their screen time on their own? What did they do?

My eldest once tried setting limits for herself because of university, but after finishing her tasks, she ended up spending a lot of time on social media again.

Interviewee 2: Abdulrahman (50)

Interviewer: Noura Altuwaim

1. What do you think about the impact of electronic devices on children's daily lives?

Devices have taken over their lives to the point where everything else seems less important.

2. What are the main factors influencing how much screen time your children get?

- a. For my youngest, it's games like Roblox.

- b. For the older kids, it's staying in touch with friends.

3. Tell me about the last time you felt worried about your children's use of electronic devices. What happened?

The last time was when I saw my youngest playing Roblox late at night. She was extremely exhausted at school the next day.

4. When do you prefer to limit your children's screen time? And why?

. During family time and prayer time because those are important moments for connection and focus.

5. Can you describe a situation where you faced a challenge in limiting your children's screen time? How did you handle it?
When I tried banning devices during dinner, my eldest objected because she was on a call with her friends. I had to allow her extra time.
6. What changes have you noticed in your children's behaviour when they spend too much time on screens? Can you share a specific example?
My youngest gets irritable when we ask her to stop playing, sometimes even crying.
7. Have you ever used an app to manage screen time? How was your experience?
No, but I'm considering using one if it's simple and effective.
8. If you had an app to help manage screen time, what feature would be most useful to you? And why?
Scheduling playtime and automatically locking devices without arguments.
9. What changes have you noticed in your children's sleep after using electronic devices? Can you share a specific example?
Their bedtime has gotten much later, especially my youngest, who sometimes stays up playing until very late.
10. Have you ever seen your child try to reduce their screen time on their own? What did they do?
My eldest tried setting limits for herself due to school pressure, but as soon as she finished her homework, she went back to using her device.

Interviewee 3: Shahad (29)

Interviewer: Alanoud Alsanad

1. What do you think about the impact of electronic devices on children's daily lives?
I believe they can be valuable when used responsibly. but excessive use can be harmful and affect their activity and daily life.
2. What are the main factors influencing how much screen time your children get?
It depends on how busy they are with schoolwork, and sometimes their friends' influence when they're online all the time.
3. Tell me about the last time you felt worried about your children's use of electronic devices. What happened?
I once noticed my son staying up late watching videos on his device. The next day, he couldn't wake up for school and was exhausted the entire day.
4. When do you prefer to limit your children's screen time? And why?
During mealtimes and before bed. I told them that life is what happens when you're busy

making other plans, I believe it's important for them to be present in the moment, especially with family.

5. Can you describe a situation where you faced a challenge in limiting your children's screen time? How did you handle it?

My daughter was stubborn and didn't want to stop playing a game, so I had to make a deal with her, if she stopped now, I'd give her extra time the next day.

6. What changes have you noticed in your children's behaviour when they spend too much time on screens? Can you share a specific example?

After a long weekend where my son played for hours, I noticed he became easily upset over the smallest things.

7. Have you ever used an app to manage screen time? How was your experience?

Yes, I tried one, but I had to unlock it for them under the excuse of studying. Over time, I learned their daily usage patterns and figured out how to manage it properly.

8. If you had an app to help manage screen time, what feature would be most useful to you? And why?

I'd like an option that locks certain apps after a set time.

9. What changes have you noticed in your children's sleep after using electronic devices? Can you share a specific example?

Yes, when they use their phones before bed, they stay up way too late and wake up exhausted. One night, I walked in on them in the middle of the night, and they were still playing, completely unaware of the time!

10. Have you ever seen your child try to reduce their screen time on their own? What did they do?

Rarely, but once, when he had exams, he handed me his phone so he could focus.

Interviewee 4: Nawaf (32)

Interviewer: Alanoud Alsanad

1. What do you think about the impact of electronic devices on children's daily lives?

Devices have their benefits, but they become harmful when overused, especially when they take time away from other activities like playing or studying.

2. What are the main factors influencing how much screen time your children get?

It depends on their tasks for the day, the number of activities they have, and sometimes even whether their friends are online.

3. Tell me about the last time you felt worried about your children's use of electronic devices.

What happened?

My son once played a game all night and woke up late for school. He was in a bad mood the entire day.

4. When do you prefer to limit your children's screen time? And why?

During study time and family time because these moments require focus and direct interaction without distractions.

5. Can you describe a situation where you faced a challenge in limiting your children's screen time? How did you handle it?

My daughter was watching YouTube for hours and refused to stop, so I had to turn off the Wi-Fi to put an end to it.

6. What changes have you noticed in your children's behaviour when they spend too much time on screens? Can you share a specific example?

When they overuse devices, they become irritable and impatient. For example, after my son spent an entire day gaming, he started yelling when I asked him for something simple.

7. Have you ever used an app to manage screen time? How was your experience?

Yes, I tried one before. It was decent, but it didn't give me the level of control I needed for each individual app.

8. If you had an app to help manage screen time, what feature would be most useful to you? And why?

I need a feature that provides detailed reports on device usage and allows me to control everything remotely without issues.

9. What changes have you noticed in your children's sleep after using electronic devices? Can you share a specific example?

They stay up much later when they use devices at night. Once, my daughter stayed up watching cartoons until midnight and couldn't wake up early the next morning.

10. Have you ever seen your child try to reduce their screen time on their own? What did they do?

When my son's school exams were approaching, he chose to put his devices aside and focus on studying, saying he would continue using them after the exams.

Interviewee 5: Mohammad (53)

Interviewer: Noura Almuayli

1. What do you think about the impact of electronic devices on children's daily lives?

Honestly, I see it as a double-edged sword. It has benefits like learning and entertainment, but at the same time, I feel like it has made them less active and less engaged with family.

2. What are the main factors influencing how much screen time your children get?

The biggest factor is free time, especially during vacations. Then comes new games and videos that spark their curiosity, making them spend even more time on screens.

3. Tell me about the last time you felt worried about your children's use of electronic devices. What happened?

Children now a days feel like screens are part of their personalities. One time, I left the house while they were on their devices, and when I came back, they were still in the same position. That's when I started feeling like it was an unhealthy addiction.

4. When do you prefer to limit your children's screen time? And why?

During meals and family gatherings, I prefer having real conversations instead of everyone being glued to their phones.

5. Can you describe a situation where you faced a challenge in limiting your children's screen time? How did you handle it?

Recently, they got hooked on a game, playing it even while walking. So, we agreed on specific times when they wouldn't use their phones.

6. What changes have you noticed in your children's behaviour when they spend too much time on screens? Can you share a specific example?

They become more irritable and anxious when I ask them to stop. Sometimes, they get so absorbed that they forget everything around them.

7. Have you ever used an app to manage screen time? How was your experience?

Yes, I tried one, but honestly, the kids are smart they found a way to bypass it! I realized that open discussions and clear rules work better.

8. If you had an app to help manage screen time, what feature would be most useful to you? And why?

I'd love an option that automatically locks certain apps after a set time without me having to intervene so there's no argument every time.

9. What changes have you noticed in your children's sleep after using electronic devices? Can you share a specific example?

When they use their phones before bed, they stay up too late and wake up exhausted. One night, I walked in at midnight and found them still playing, completely unaware of the time!

10. Have you ever seen your child try to reduce their screen time on their own? What did they do?

Rarely, but once, my son saw his friend cutting back on phone use and playing more outside. He tried doing the same and slightly reduced his screen time.

Interviewee 6: Muneera (45)

Interviewer: Noura Almuayli

1. What do you think about the impact of electronic devices on children's daily lives?
I think they've become a huge part of their lives, but at the expense of other things, like playing with family or even reading books.
2. What are the main factors influencing how much screen time your children get?
Friends and daily habits, if everyone around them is always on their devices, it's naturally harder to convince them to cut back.
3. Tell me about the last time you felt worried about your children's use of electronic devices. What happened?
A while ago, I asked my daughter to help me with something, but she was so glued to her phone that she didn't even hear me! I called her multiple times, and she only noticed when I raised my voice. That's when I realized it was getting out of hand.
4. When do you prefer to limit your children's screen time? And why?
Before bedtime, because if they use screens late at night, they stay up too long, and in the morning, they look like they barely slept.
5. Can you describe a situation where you faced a challenge in limiting your children's screen time? How did you handle it?
Yes, my daughter was addicted to Snapchat, and every time I tried to limit her time on it, she felt like I was invading her privacy.
6. What changes have you noticed in your children's behaviour when they spend too much time on screens? Can you share a specific example?
They become easily distracted and constantly forget their responsibilities. One time, my son said he would do his homework after finishing a game, and the whole day passed without him doing it.
7. Have you ever used an app to manage screen time? How was your experience?
No, but I think an app like that would be useful.
8. If you had an app to help manage screen time, what feature would be most useful to you? And why?
I'd love a daily schedule that's easy to adjust, so it can automatically set screen time limits without me having to control it manually every day.
9. What changes have you noticed in your children's sleep after using electronic devices? Can you share a specific example?
If they use their devices before bed, they end up staying up late, and their sleep becomes disrupted.
10. Have you ever seen your child try to reduce their screen time on their own? What did they do?
Yes, my daughter decided to cut down on phone use after she started experiencing frequent headaches.

Appendix B: Questionnaires

1. Personal Information

1. عمر ولي الأمر ؟
2. جنس ولي الأمر ؟
3. كم عدد أطفالك ؟
4. أعمار أطفالك (اختر جميع ما ينطبق) ؟

2. Close-ended Questions

1. هل تواجه صعوبة في التحكم بوقت استخدام أطفالك للأجهزة ؟
2. هل تشعر بالقلق في عدم القدرة على معرفة التطبيقات التي يقضيها طفلك على الجهاز ؟
3. كم عدد الساعات التي يقضيها أطفالك على الأجهزة يوميًا (تقريبًا) ؟
4. ما أكثر الأنشطة التي يقوم بها أطفالك على الأجهزة ؟
5. ما هي أكثر الأوقات التي ترغب في تقليل استخدام أطفالك للأجهزة فيها (اختر جميع ما ينطبق) ؟
6. ماهي التحديات التي تواجهك في تنظيم وقت الشاشة ؟
7. ما الميزات التي تجدها ضرورية في التطبيق لتنظيم وقت الشاشة ؟
8. هل ترى أن وجود ميزة تربط وقت الشاشة بمهام معينة (مثل إنهاء الواجبات) سيكون مفيدًا ؟

Distributed using Google Forms via WhatsApp to the targeted groups.

Link to the raw data file which show individual responses: [Google Form Link](#)

Link to the raw data file which show individual responses: [Excel Link](#)

Appendix C: Satisfaction Questionnaires

• Post-Task Questions

1. The task was easy to complete.
2. I didn't feel confused or lost while performing this task.
3. The task flow was smooth and intuitive to follow.
4. The buttons and their functions in this task were clear and easy to understand.
5. I'm satisfied with the results I got after completing this task.

• Post-Test Questions

1. The test was easy to complete.
2. I clearly understood what I was supposed to do in this test.
3. The layout and design made it easy for me to complete tasks efficiently.
4. The system's interface felt familiar and intuitive to use.
5. I would be comfortable using this system again in the future.